Chapter 2

Spatial Descriptions and Transformations

Pengcheng Hu, SMMG, HKUST(GZ)

pengchenghu@hkust-gz.edu.cn

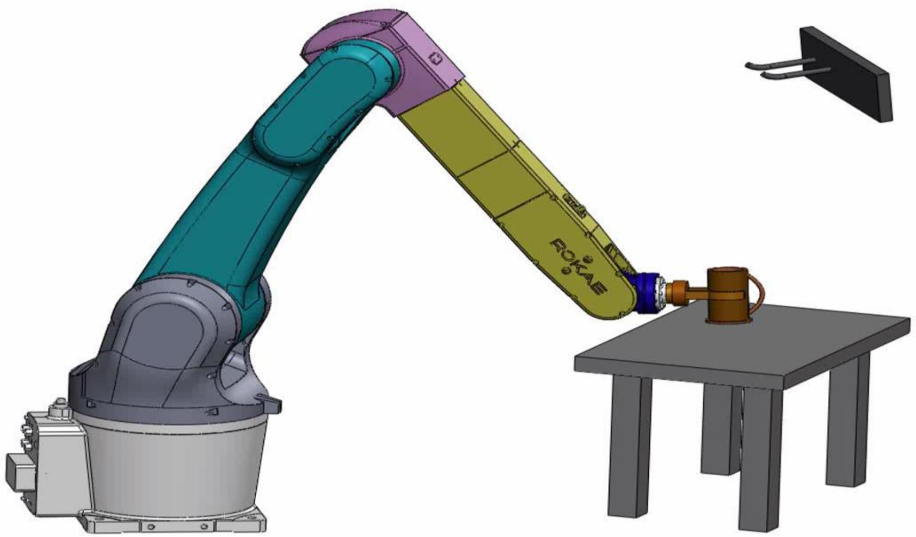
Chapter 2 Spatial Descriptions and Transformations

1. Introduction
2. Description of Translation and Rotation
3. Rotation Matrix
4. Other Methods of Posture Description
5. Integration of Translation and Rotation
6. Homogeneous Transformation Matrix

1. Introduction

Robotic manipulation implies that parts and tools will be moved around in space by some sort of mechanism - naturally leads to a **need for representing positions and the orientations of parts**, of tools, and of the mechanism itself.

- How to describe the position and orientation of an object-cup in space?
- How to describe the position and orientation in space of the joints of the manipulator arm?

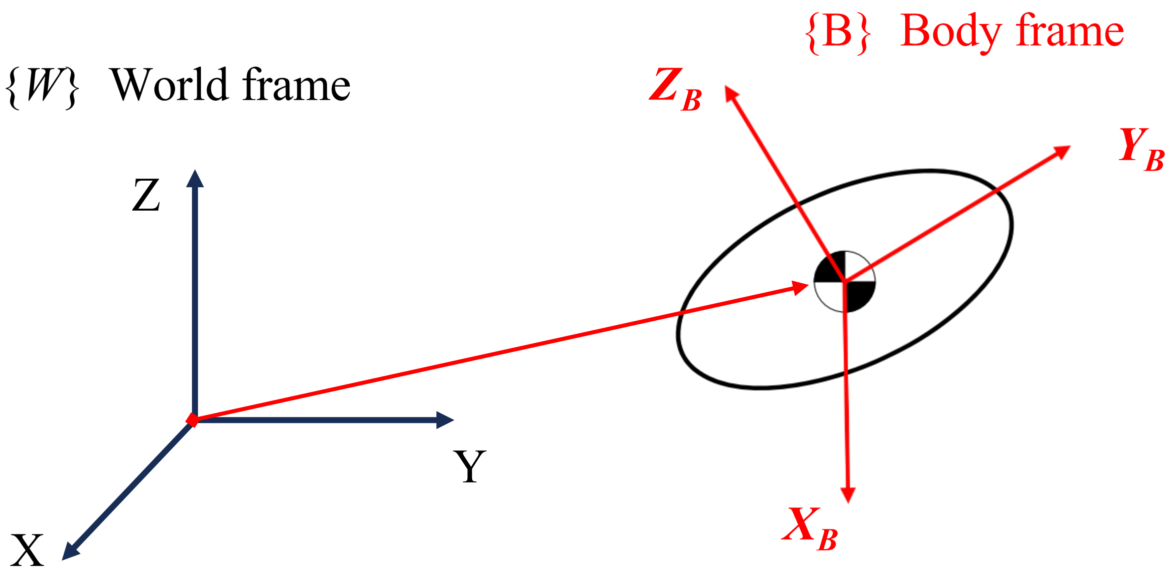


Task: how to pick up a cup on the table and hang it on the wall-mounted hook?

1. Introduction

- How to describe the state of a **rigid body**?
  - 2D Plane – 2 Translation Degrees of Freedom (DOFs) and 1 Rotation DOF
  - 3D Space – 3 Translation DOFs and 3 Rotation DOFs

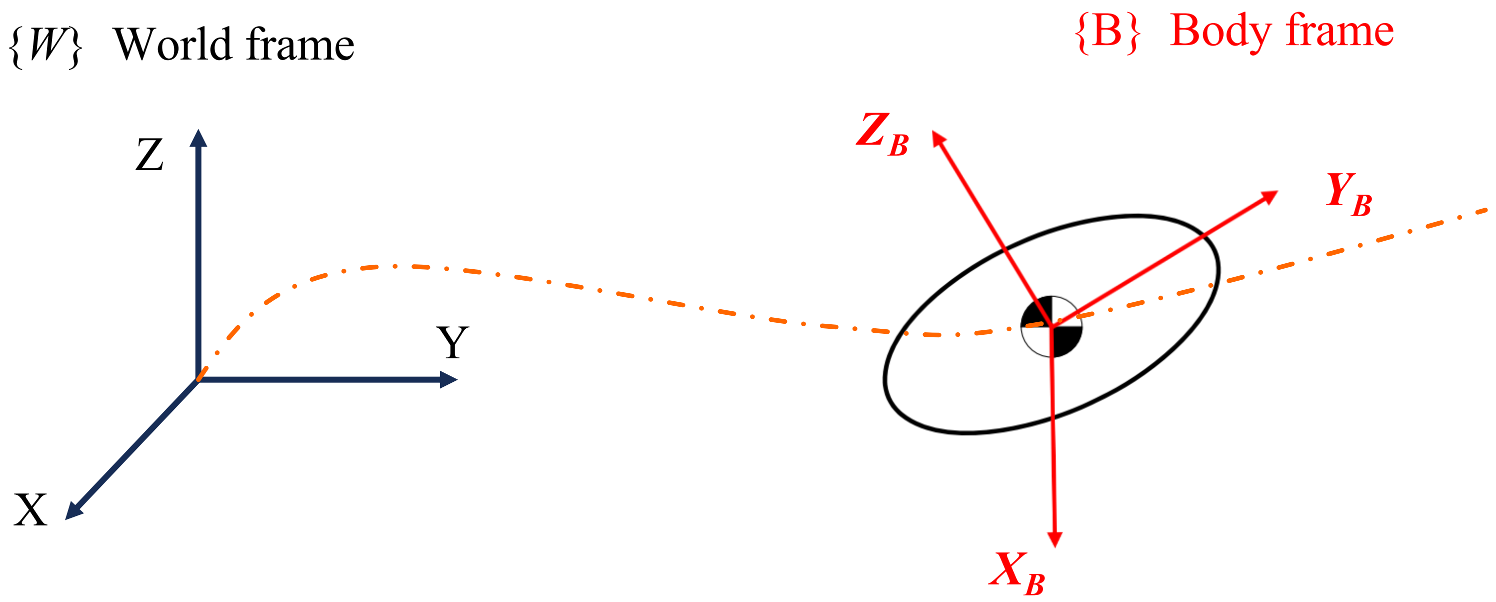
- How to integrate the expression of the state of the rigid body?
  - Create a frame on a rigid body, often on the center of mass
    - ✓ Translation: determined by **the position of origin** of the body frame.
    - ✓ Rotate: determined by **the posture** of the body frame.



1. Introduction

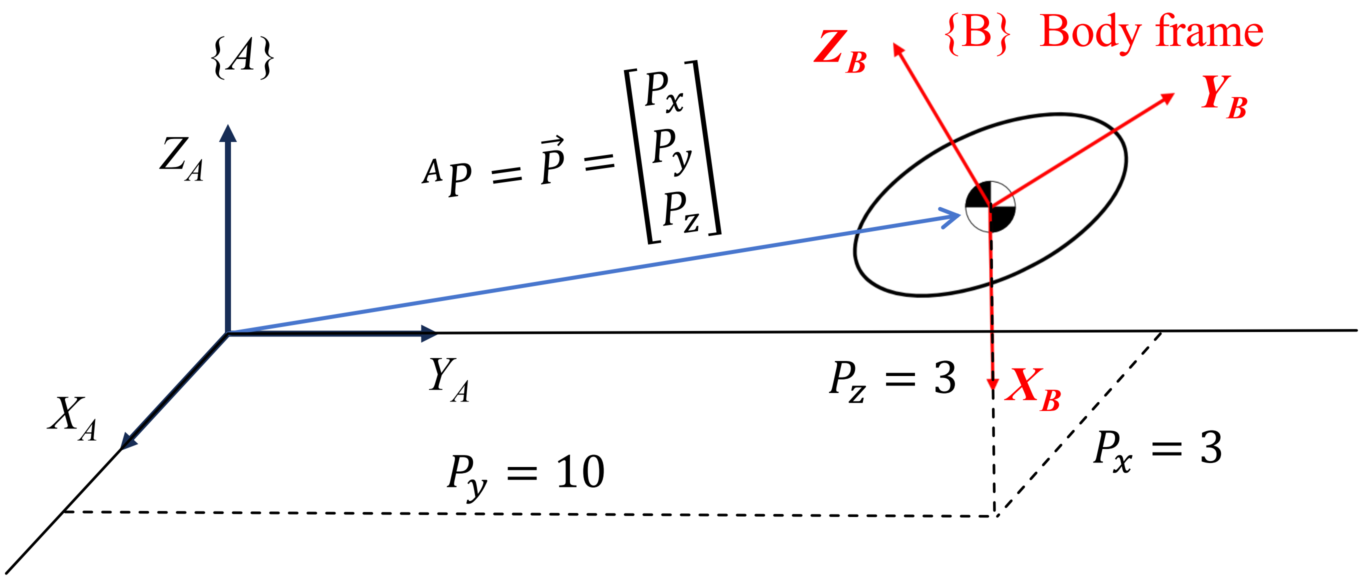
- How to describe the "motion" of a rigid body (frame)?

  - Conversion of displacement and orientation to velocity and acceleration using differentiation of individual DOFs.



2. Description of Translation and Rotation

□ **Translation** : use vector $\vec{P}$ to describe the state of the origin of {B} with respect to {A}.



$$^{A}P = \vec{P} = \begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \text{Origin of \{B\} represented in \{A\}}$$

Vectors are written with a **leading superscript** indicating the coordinate system to which they are referenced: for example, ^{A}P.

2. Description of Translation and Rotation

☐ **Rotation**: describes the posture of {B} relative to {A} – Rotation Matrix

$$ {}^{A}_{B}R = \begin{bmatrix} | & | & | \\ {}^{A}X_B & {}^{A}Y_B & \boxed{{}^{A}Z_B} \\ | & | & | \end{bmatrix} $$

The three columns of R are the principle direction unit vector of {B}: X_B , Y_B , Z_B (in {A})

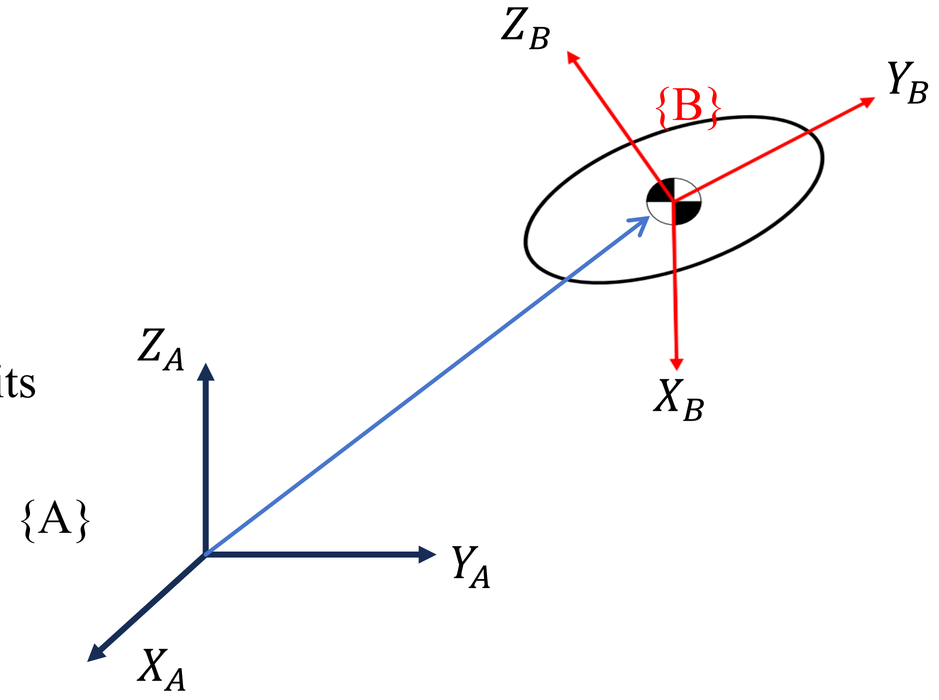
"column vector"

B relative to A

$$ = \begin{bmatrix} X_B \cdot X_A & Y_B \cdot X_A & Z_B \cdot X_A \\ X_B \cdot Y_A & Y_B \cdot Y_A & Z_B \cdot Y_A \\ X_B \cdot Z_A & Y_B \cdot Z_A & Z_B \cdot Z_A \end{bmatrix} $$

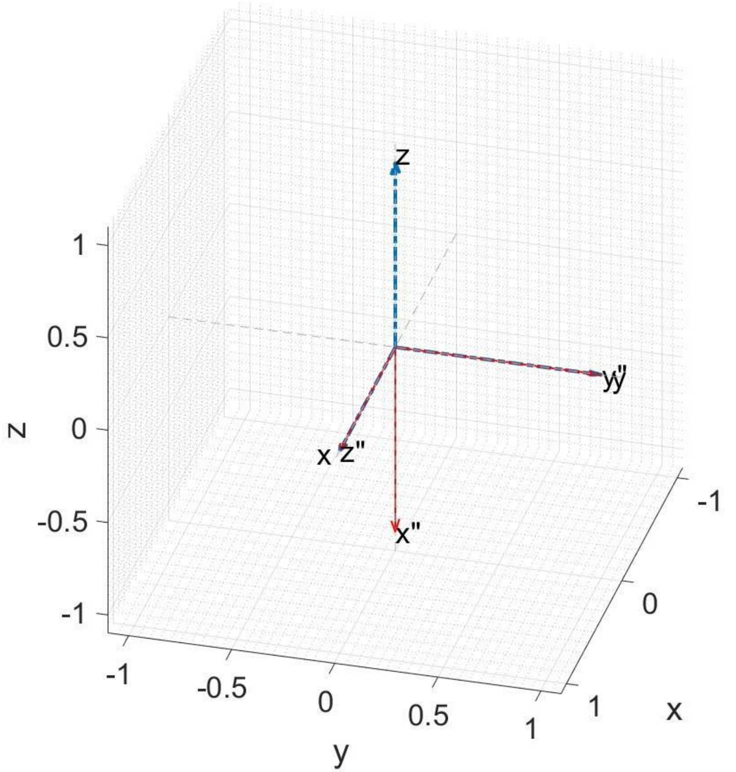
Projections of X_B, Y_B and Z_B onto the unit directions of its reference frame - "**direction cosines**"

The rotation matrix ${}^{A}_{B}R$ is a 3×3 matrix.



2. Description of Translation and Rotation

- **Ex**: Posture of {B} with respect to {A} ${}^{A}_{B}R = ?$



Blue dotted line: World frame {A}

Red solid line: Body frame {B}

The X'' of {B} is the reverse of the Z-axis of {A}: ${}^{A}X_B = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$

The Y'' of {B} overlaps the Y-axis of {A}: ${}^{A}Y_B = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$

The Z'' of {B} overlaps the X-axis of {A}: ${}^{A}Z_B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$

Therefore, the posture of {B} with respect to {A} :

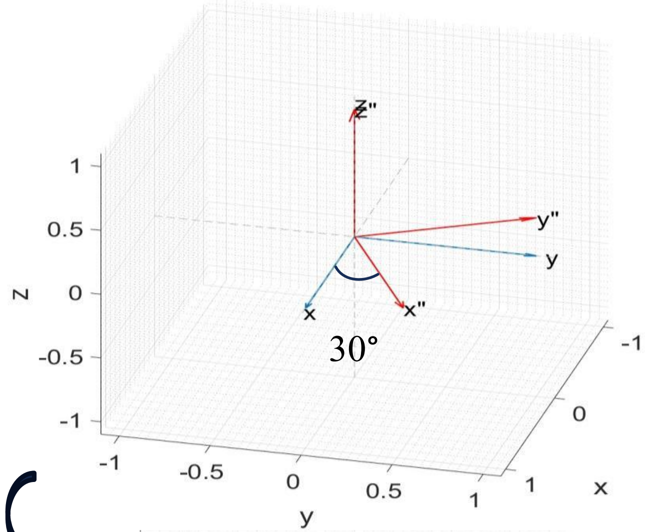
$$
{}^{A}_{B}R = \begin{bmatrix} | & | & | \\ {}^{A}X_B & {}^{A}Y_B & {}^{A}Z_B \\ | & | & | \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix}
$$

2. Description of Translation and Rotation

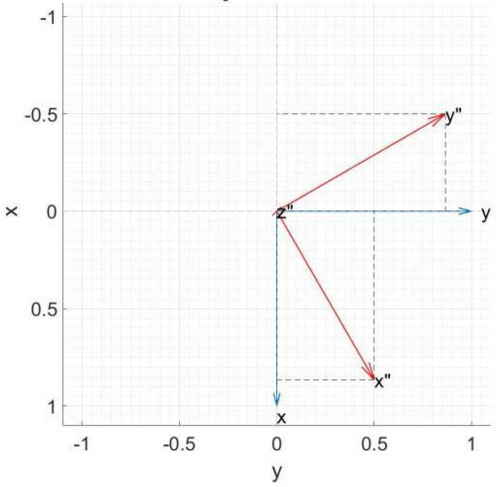
- **Ex**: Posture of {B} with respect to {A} ${}^A_BR = ?$

Blue dotted line: World frame {A}

Red solid line: Body frame {B}



Top view



$$ {}^AX_B = \begin{bmatrix} X_B \cdot X_A \\ X_B \cdot Y_A \\ X_B \cdot Z_A \end{bmatrix} = \begin{bmatrix} 0.866 \\ 0.5 \\ 0 \end{bmatrix} $$

$$ {}^AY_B = \begin{bmatrix} Y_B \cdot X_A \\ Y_B \cdot Y_A \\ Y_B \cdot Z_A \end{bmatrix} = \begin{bmatrix} -0.5 \\ 0.866 \\ 0 \end{bmatrix} $$

$$ {}^AZ_B = \begin{bmatrix} Z_B \cdot X_A \\ Z_B \cdot Y_A \\ Z_B \cdot Z_A \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} $$

Posture of {B} with respect to {A}:

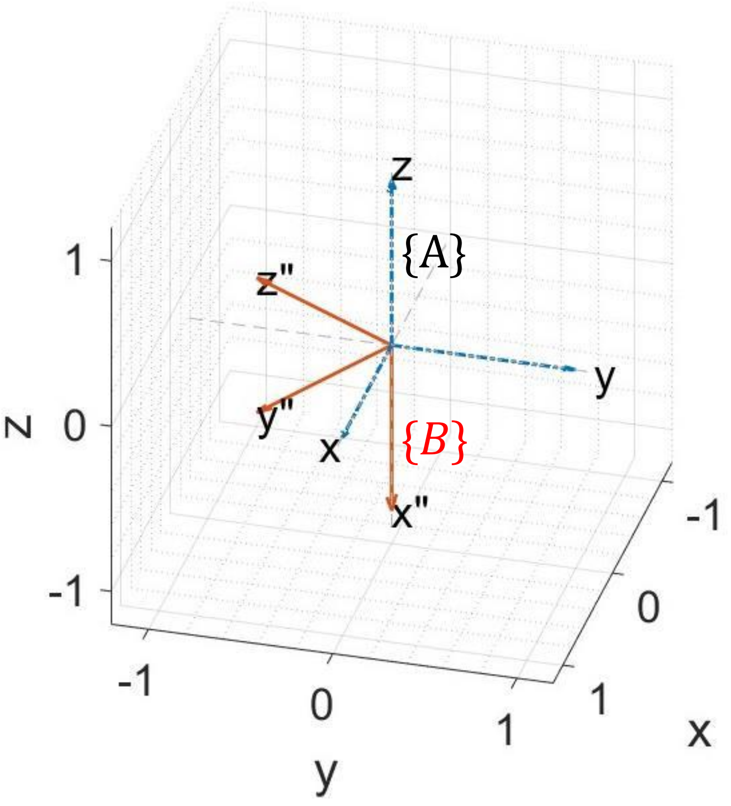
$$ {}^A_BR = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} $$

2. Description of Translation and Rotation

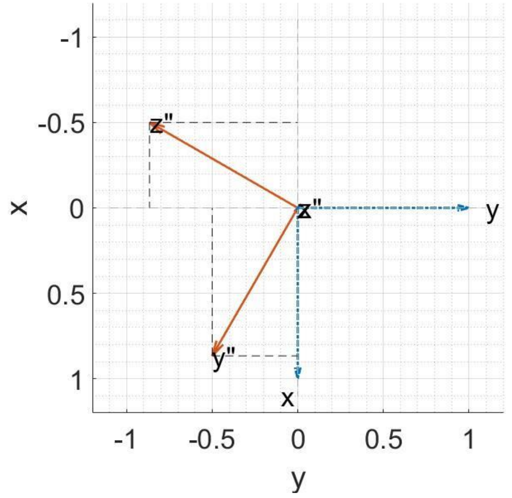
☐ **Quiz**: Posture of {B} with respect to {A} ${}^{A}_{B}R = ?$

Blue dotted line: World frame {A}

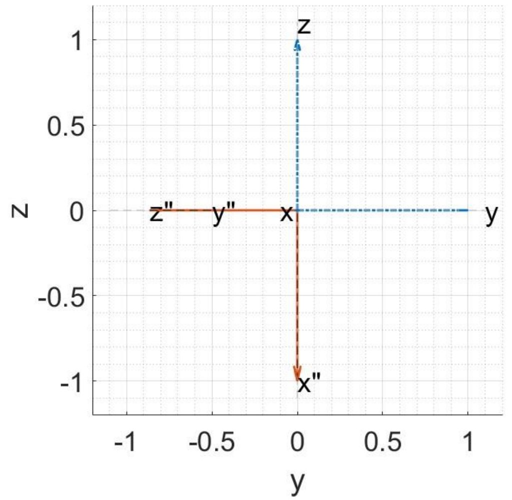
Red solid line: Body frame {B}



XY plane



ZY plane



A. $\begin{bmatrix} 0 & -0.866 & 0.5 \\ 0 & 0.5 & 0.866 \\ -1 & 0 & 0 \end{bmatrix}$

B. $\begin{bmatrix} 0 & 0.866 & -0.5 \\ 0 & -0.5 & -0.866 \\ -1 & 0 & 0 \end{bmatrix}$

C. $\begin{bmatrix} 0 & 0 & -1 \\ 0.866 & -0.5 & 0 \\ -0.5 & -0.866 & 0 \end{bmatrix}$

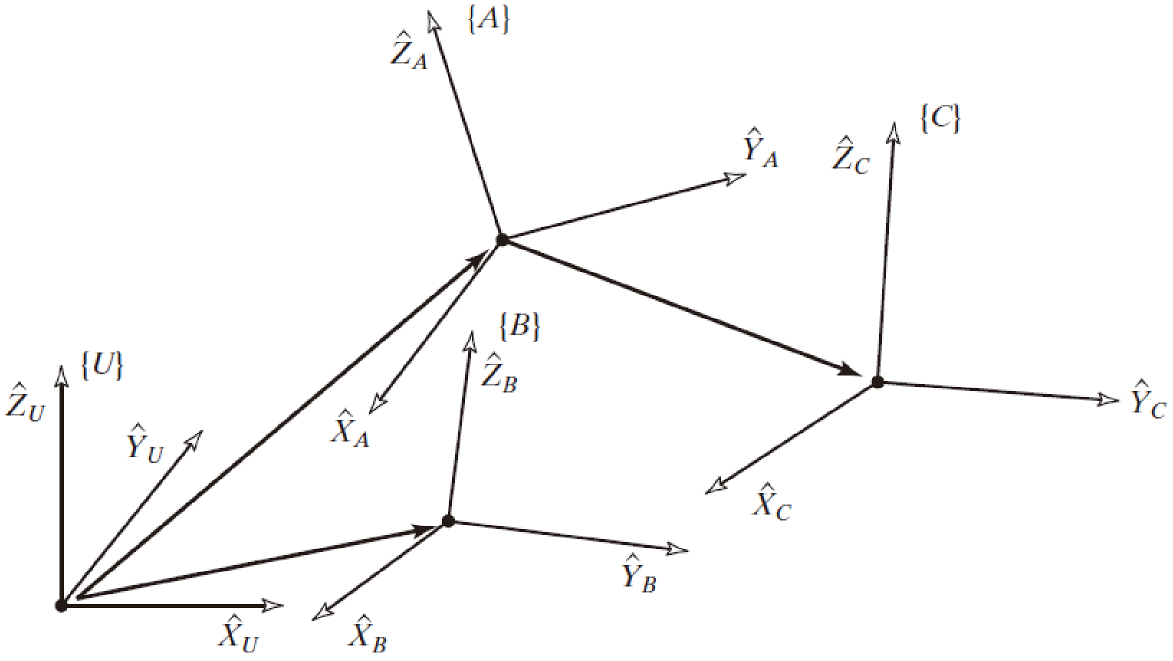
D. $\begin{bmatrix} 0 & 0 & -1 \\ -0.866 & 0.5 & 0 \\ 0.5 & 0.866 & 0 \end{bmatrix}$

2. Description of Translation and Rotation

- **Frame**: a set of four vectors giving position and orientation information

Frame $\{B\}$ is described by ${}_{B}^{A}R$ and ${}^{A}P_{BORG}$, and ${}^{A}P_{BORG}$ is the vector that locates the origin of the frame $\{B\}$:

$$\{B\}=\{{}_{B}^{A}R,\ {}^{A}P_{BORG}\}$$

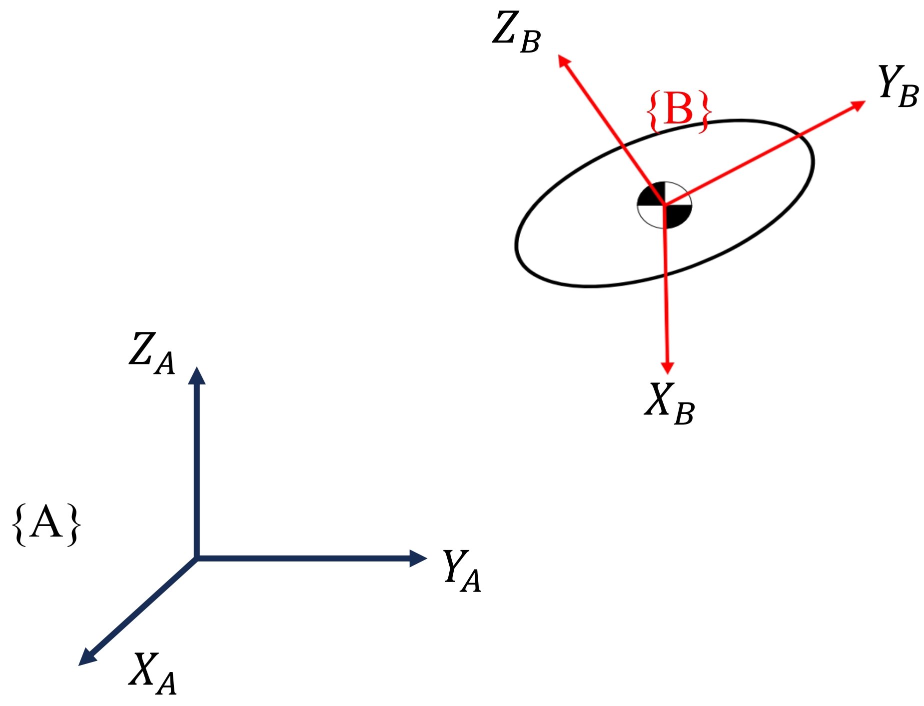


Example of several frames: Frames $\{A\}$ and $\{B\}$ are known relative to the universe coordinate system $\{U\}$, frame $\{C\}$ is known relative to frame $\{A\}$.

3. Rotation Matrix

- Rotation matrix describes $\{B\}$relative to $\{A\}$, name it with the notation ${}^A_BR$
- Characteristics of ${}^A_BR$

$$
{}^A_BR = \begin{bmatrix} X_B \cdot X_A & Y_B \cdot X_A & Z_B \cdot X_A \\ X_B \cdot Y_A & Y_B \cdot Y_A & Z_B \cdot Y_A \\ X_B \cdot Z_A & Y_B \cdot Z_A & Z_B \cdot Z_A \end{bmatrix} = \begin{bmatrix} | & | & | \\ {}^AX_B & {}^AY_B & {}^AZ_B \\ | & | & | \end{bmatrix}
$$



Forward and backward vector swaps

$$
= \begin{bmatrix} X_A \cdot X_B & X_A \cdot Y_B & X_A \cdot Z_B \\ Y_A \cdot X_B & Y_A \cdot Y_B & Y_A \cdot Z_B \\ Z_A \cdot X_B & Z \cdot Y_B & Z_A \cdot Z_B \end{bmatrix} = \begin{bmatrix} - & {}^BX_A^T & - \\ - & {}^BY_A^T & - \\ - & {}^BZ_A^T & - \end{bmatrix}
$$

$$
= \begin{bmatrix} | & | & | \\ {}^BX_A & {}^BY_A & {}^BZ_A \\ | & | & | \end{bmatrix}^T = {}^B_AR^T
$$

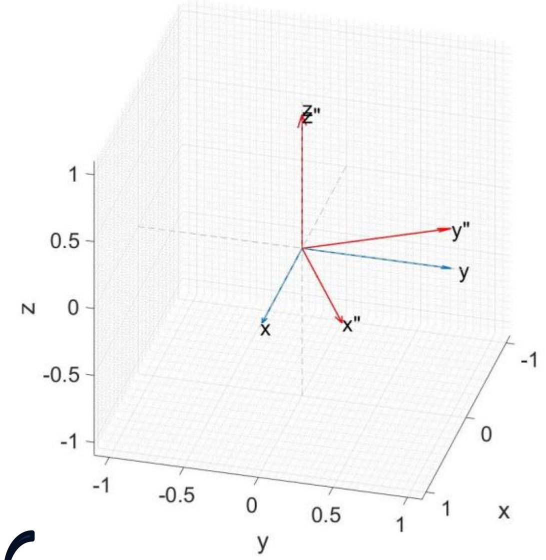
$$
\Rightarrow \quad {}^A_BR = {}^B_AR^{-1} = {}^B_AR^T \qquad \text{i.e., } {}^A_BR^T\, {}^A_BR = I_3
$$

Linear algebra: **inverse of an orthogonal matrix equals to its transpose**

3. Rotation Matrix

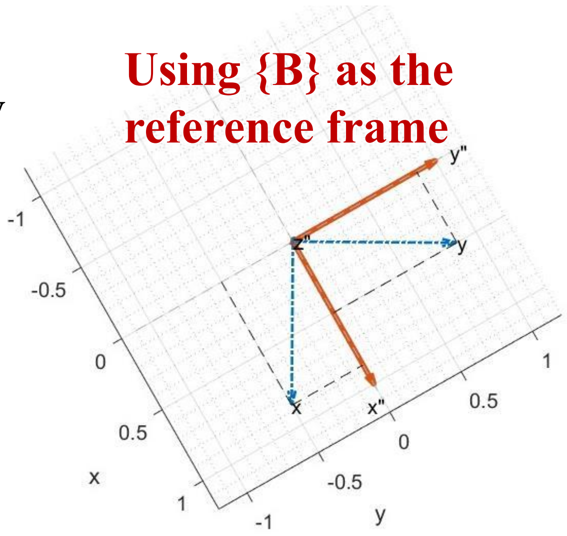
- **Ex**: Posture of {A} with respect to {B} ${}^{B}_{A}R = ?$

$$
{}^{A}_{B}R = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix}
$$



Top view

Using {B} as the reference frame



Blue dotted line: World frame {A}

Red solid line: Body frame {B}

$$
{}^{B}X_A = \begin{bmatrix} X_A \cdot X_B \\ X_A \cdot Y_B \\ X_A \cdot Z_B \end{bmatrix} = \begin{bmatrix} 0.866 \\ -0.5 \\ 0 \end{bmatrix}
$$

$$
{}^{B}Y_A = \begin{bmatrix} Y_A \cdot X_B \\ Y_A \cdot Y_B \\ Y_A \cdot Z_B \end{bmatrix} = \begin{bmatrix} 0.5 \\ 0.866 \\ 0 \end{bmatrix}
$$

$$
{}^{B}Z_A = \begin{bmatrix} Z_A \cdot X_B \\ Z_A \cdot Y_B \\ Z_A \cdot Z_B \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}
$$

Posture of {A} with respect to {B}:

$$
{}^{B}_{A}R = \begin{bmatrix} 0.866 & 0.5 & 0 \\ -0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} = {}^{A}_{B}R^{T}
$$

The result of the previous example

3. Rotation Matrix

- A 3 × 3 orthogonal matrix Q $QQ^T = Q^T Q = I$

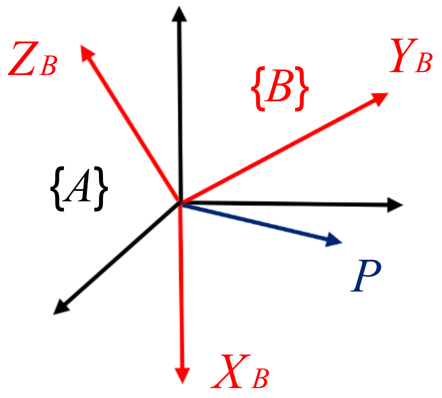
- Always invertible $Q^{-1} = Q^T$
- Columns: orthonormal basis
  - Length (Norm) = 1
  - Mutually perpendicular

$$
{}^{A}_{B}R = \begin{bmatrix} X_B \cdot X_A & Y_B \cdot X_A & Z_B \cdot X_A \\ X_B \cdot Y_A & Y_B \cdot Y_A & Z_B \cdot Y_A \\ X_B \cdot Z_A & Y_B \cdot Z_A & Z_B \cdot Z_A \end{bmatrix}
$$

- The Rotation matrix (R) has 9 numbers, but the two conditions in the above column place 6 constraints, so R has only 3 DOFs, which is consistent with a rotation in space having 3 DOFs
- Determinant = 1 (rotation); = -1 (reflection)

3. Rotation Matrix

- The Rotation Matrix, in addition to describing the posture of $\{B\}$ with respect to $\{A\}$, can also be used to transform the coordinates of the **vectors**.



Original coordinate: ${}^{B}P = {}^{B}P_x X_B + {}^{B}P_y Y_B + {}^{B}P_z Z_B$

New coordinate: ${}^{A}P = {}^{A}P_x X_A + {}^{A}P_y Y_A + {}^{A}P_z Z_A$

Where

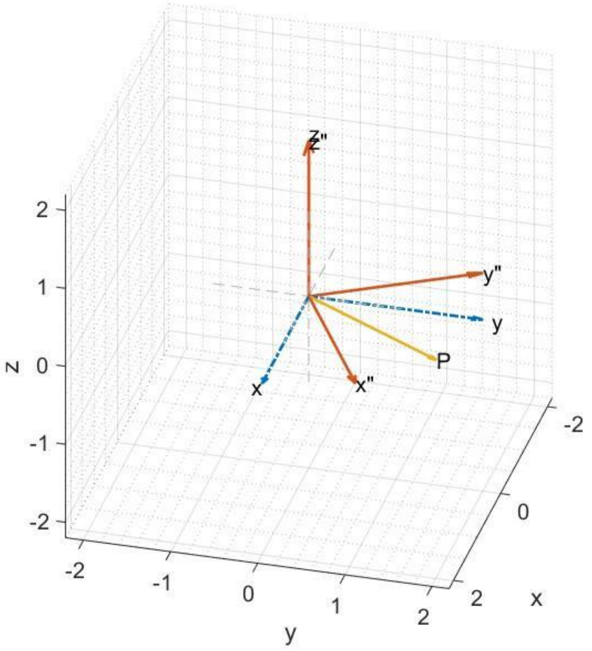
$$
\begin{aligned}
{}^{A}P_x &= {}^{B}P \cdot X_A = X_B \cdot X_A\, {}^{B}P_x + Y_B \cdot X_A\, {}^{B}P_y + Z_B \cdot X_A\, {}^{B}P_z \\
{}^{A}P_y &= {}^{B}P \cdot Y_A = X_B \cdot Y_A\, {}^{B}P_x + Y_B \cdot Y_A\, {}^{B}P_y + Z_B \cdot Y_A\, {}^{B}P_z \\
{}^{A}P_z &= {}^{B}P \cdot Z_A = X_B \cdot Z_A\, {}^{B}P_x + Y_B \cdot Z_A\, {}^{B}P_y + Z_B \cdot Z_A\, {}^{B}P_z
\end{aligned}
$$

$$
\Rightarrow \quad {}^{A}P = {}^{A}\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = \begin{bmatrix} X_B \cdot X_A & Y_B \cdot X_A & Z_B \cdot X_A \\ X_B \cdot Y_A & Y_B \cdot Y_A & Z_B \cdot Y_A \\ X_B \cdot Z_A & Y_B \cdot Z_A & Z_B \cdot Z_A \end{bmatrix} {}^{B}\begin{bmatrix} P_x \\ P_y \\ P_z \end{bmatrix} = {}^{A}_{B}R\, {}^{B}P
$$

3. Rotation Matrix

- **Ex**: If the relative states of $\{B\}$ and $\{A\}$ are the same as shown in "Rotation" *P9*, suppose

  ${}^{B}P = [1.732\ 1\ 0]^T$, ${}^{A}P = ?$

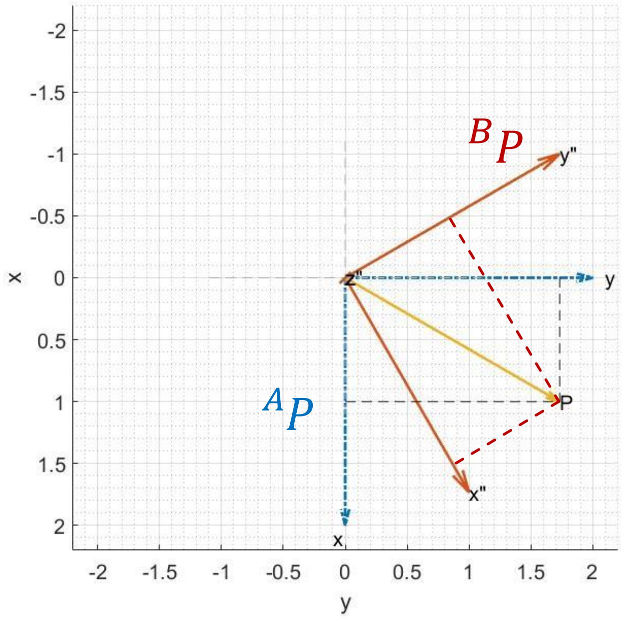


Blue dotted line: World frame {A}
Red solid line: Body frame {B}

$${}^{A}P = {}^{A}_{B}R\,{}^{B}P$$

$${}^{A}P = \begin{bmatrix} 0.866 & -0.5 & 0 \\ 0.5 & 0.866 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1.732 \\ 1 \\ 0 \end{bmatrix}$$

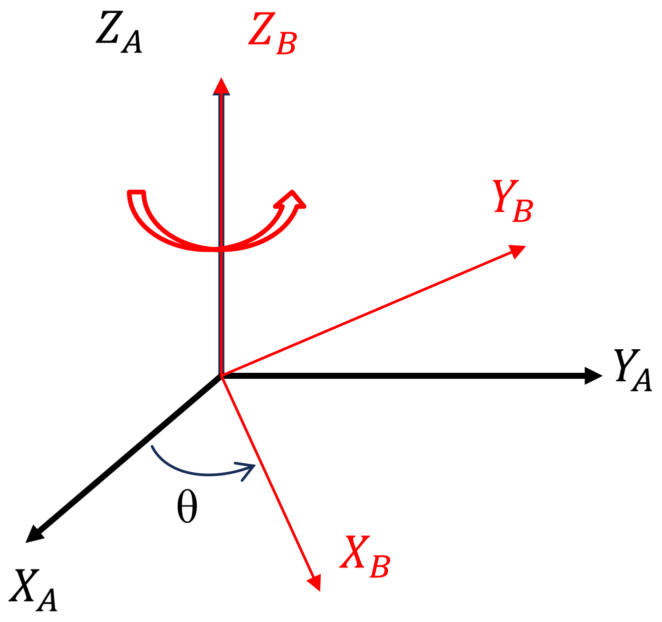
$${}^{A}P = \begin{bmatrix} 1 \\ 1.732 \\ 0 \end{bmatrix}$$



Upper view verifies the answer

3. Rotation Matrix

- The Rotation matrix can be used to characterize the state of "rotation" of an object.
- **Case1 :** Rotation along Z_A with θ - $R_{Z_A}(\theta)$



Angle of rotation

$$R_{Z_A}(\theta) = \begin{bmatrix} cos\theta & -sin\theta & 0 \\ sin\theta & cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

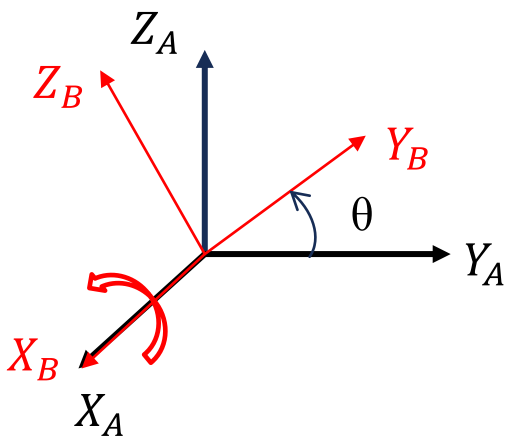
Axis of rotation

$$= \begin{bmatrix} c\theta & -s\theta & 0 \\ s\theta & c\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Note: $${}^{A}_{B}R = \begin{bmatrix} | & | & | \\ {}^{A}X_B & {}^{A}Y_B & {}^{A}Z_B \\ | & | & | \end{bmatrix}$$

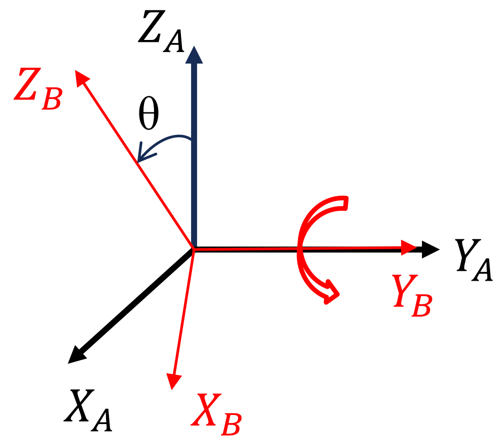
3. Rotation Matrix

- **Case 2**: Rotation along X_A with θ



$$R_{X_A}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos\theta & -sin\theta \\ 0 & sin\theta & cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & -s\theta \\ 0 & s\theta & c\theta \end{bmatrix}$$

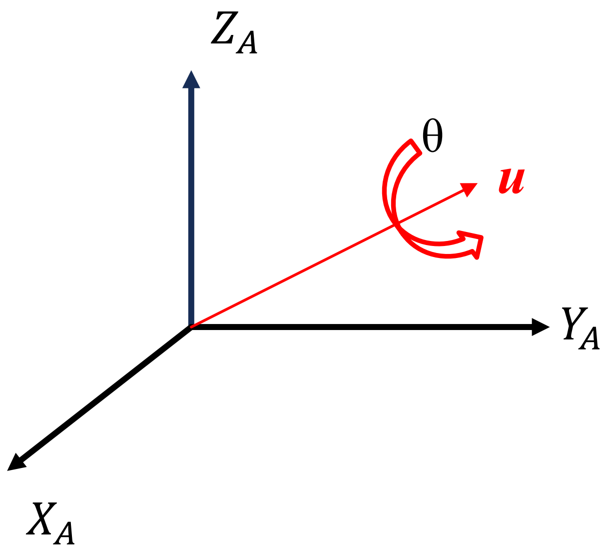
- **Case 3**: Rotation along Y_A with θ



$$R_{Y_A}(\theta) = \begin{bmatrix} cos\theta & 0 & sin\theta \\ 0 & 1 & 0 \\ -sin\theta & 0 & cos\theta \end{bmatrix} = \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix}$$

3. Rotation Matrix

- [ ] **Case 4**: Rotation along a 3D vector ***u*** = (u_x, u_y, u_z) by angle θ



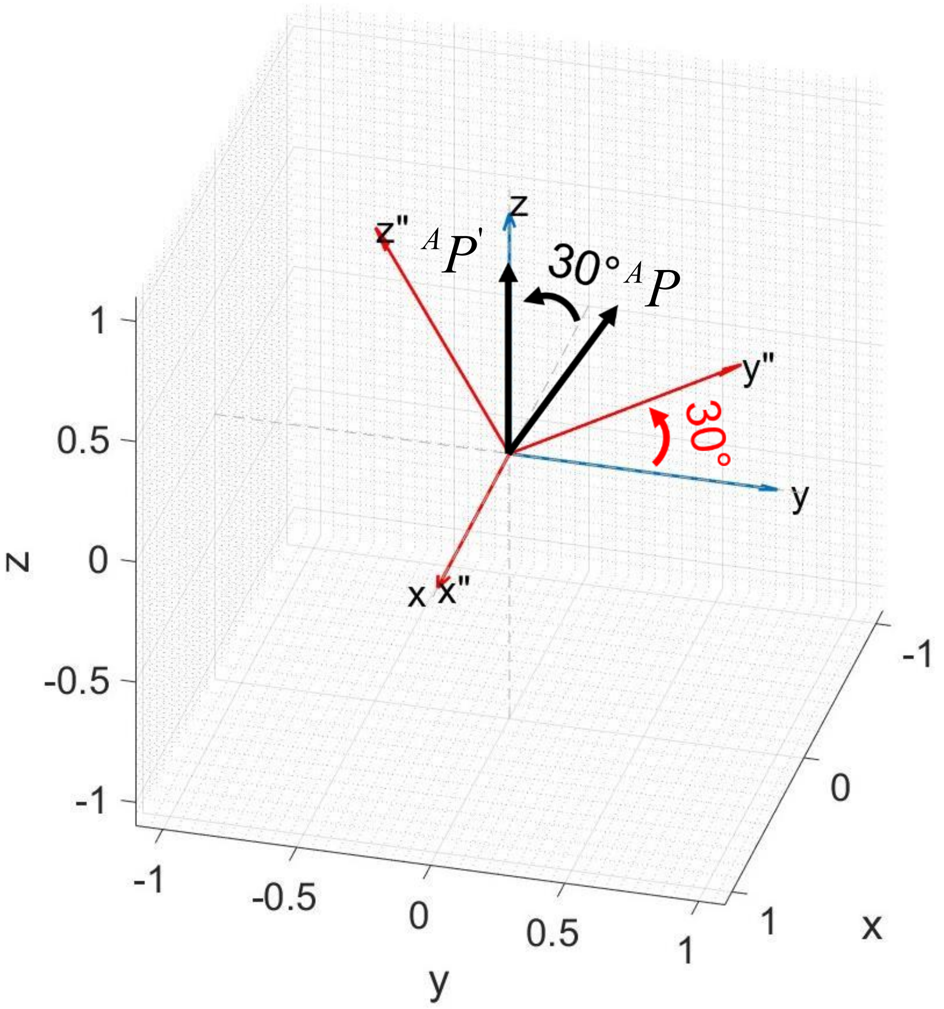
The rotation matrix ***R***:

$$R = \begin{bmatrix} u_x^2\left(1-\cos\theta\right)+\cos\theta & u_x u_y\left(1-\cos\theta\right)-u_z\sin\theta & u_x u_z\left(1-\cos\theta\right)+u_y\sin\theta \\ u_x u_y\left(1-\cos\theta\right)+u_z\sin\theta & u_y^2\left(1-\cos\theta\right)+\cos\theta & u_y u_z\left(1-\cos\theta\right)-u_x\sin\theta \\ u_x u_z\left(1-\cos\theta\right)-u_y\sin\theta & u_y u_z\left(1-\cos\theta\right)+u_x\sin\theta & u_z^2\left(1-\cos\theta\right)+\cos\theta \end{bmatrix}$$

Reference: https://en.wikipedia.org/wiki/Rotation_matrix

3. Rotation Matrix

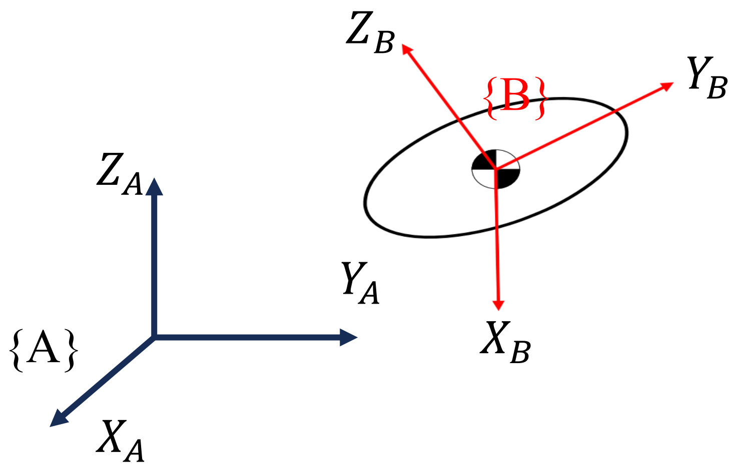
- **Ex**: ${}^{A}P = [0\ 1\ 1.732]^T$ rotates 30° around the X-axis, ${}^{A}P' = ?$



$$R_{X_A}(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & cos30° & -sin30° \\ 0 & sin30° & cos30° \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix}$$

$${}^{A}P' = R_{X_A}(\theta)\ {}^{A}P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 1.732 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 2 \end{bmatrix}$$

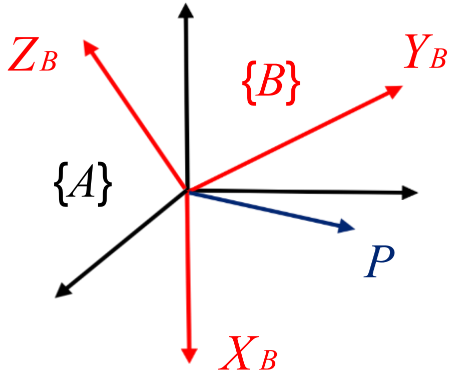
3. Rotation Matrix



- Three applications of the Rotation matrix

  - (1) Describe the posture of a frame (relative to another frame)

$$ {}^{A}_{B}R = \begin{bmatrix} | & | & | \\ {}^{A}X_B & {}^{A}Y_B & {}^{A}Z_B \\ | & | & | \end{bmatrix} $$

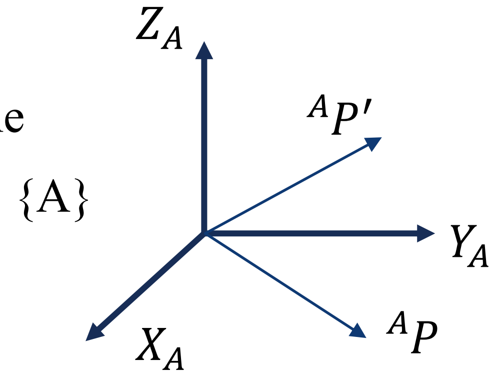


  - (2) Change the point from one frame to another which has only relative rotation to this frame

$$ {}^{A}P = {}^{A}_{B}R\,{}^{B}P $$

  - (3) Rotate the point (vector) within the same frame

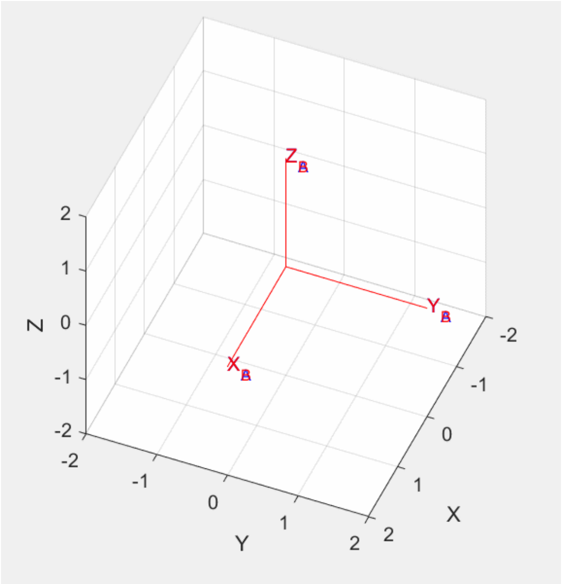
$$ {}^{A}P' = R(\theta)\ {}^{A}P $$



4. Other Posture Description

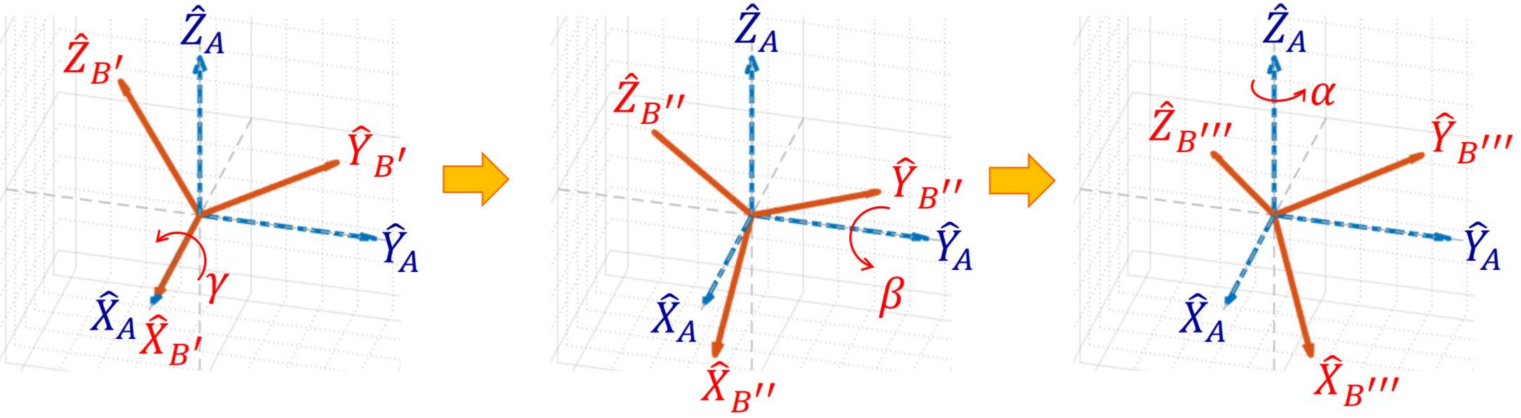
- Rotation in space has 3 DOFs; how to take the posture expressed by a general rotation matrix and break it down into 3 rotation angles to correspond to 3 DOFs?
- Notices for disassembling the rotation into a "triple-rotation multiplier"
  - Rotation is not commutable; the sequence of multiple rotations needs to be clearly defined
  - The axis of rotation also needs to be clearly defined;
    - "Fixed" axis (e.g., World Frame) rotation?
    - "Temporary rotating axis" (e.g., Body Frame) rotation?
- Two ways to disassemble
  - Rotation in the direction of "fixed axis": **Fixed angles**.
  - Rotation in the direction of "temporary rotating axis": **Euler angles**.

4. Other Posture Description: Fixed angles



X-Y-Z Fixed Angle - Deriving R from Angle

Each of the three rotations takes place about an axis in the fixed reference frame $\{A\}$ – **Fixed angle**.



$${}^A_BR_{XYZ}(\gamma,\beta,\alpha) = R_Z(\alpha)R_Y(\beta)R_X(\gamma)$$

$$= \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix}$$

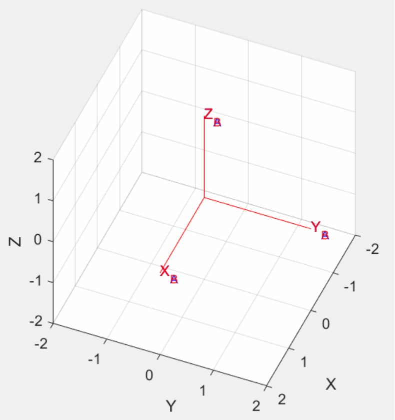
$$= \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix}$$

$$v' = {}^A_BR\, v = R_3R_2R_1v$$

Note: **first rotation puts in the "back"** , think in terms of operators, rotating or moving a vector "in the same frame".

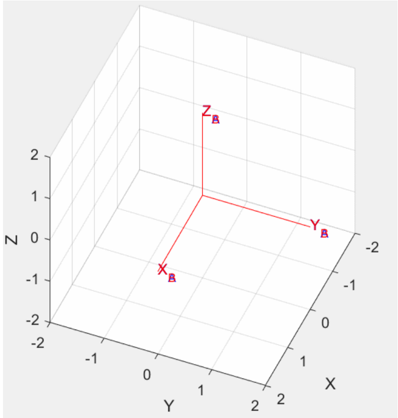
4. Other Posture Description: Fixed angles

- **Ex**: What are the respective ${}^{A}_{B}R$ when rotating with Fixed Angles?
  - (1) "Rotate the X-axis by 60 degrees first, then rotate the Y-axis by 30 degrees"
  - (2) "Rotate the Y-axis by 30 degrees first, then rotate the X-axis by 60 degrees"



- (1) Rotate the X-axis by 60 degrees first, then rotate the Y-axis by 30 degrees

$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = R_Z(0)R_Y(30)R_X(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix} $$



- (2) Rotate the Y-axis by 30 degrees first, then rotate the X-axis by 60 degrees

$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = R_Z(0)R_X(60)R_Y(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix} $$

4. Other Posture Description: Fixed angles

- X-Y-Z Fixed Angles - deriving angles from R

$$ {}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = \begin{bmatrix} c\alpha c\beta & c\alpha s\beta s\gamma - s\alpha c\gamma & c\alpha s\beta c\gamma + s\alpha s\gamma \\ s\alpha c\beta & s\alpha s\beta s\gamma + c\alpha c\gamma & s\alpha s\beta c\gamma - c\alpha s\gamma \\ -s\beta & c\beta s\gamma & c\beta c\gamma \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} $$

Case 1: If $\beta \neq 90°$:

$$\beta = Atan2(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2})$$

$$\alpha = Atan2(r_{21}/c\beta, r_{11}/c\beta)$$

$$\gamma = Atan2(r_{32}/c\beta, r_{33}/c\beta)$$

$-90° \leq \beta \leq 90°$

Single solution

Case 2: If $\beta = 90°$:

$$\alpha = 0$$

$$\gamma = Atan2(r_{12}, r_{22})$$

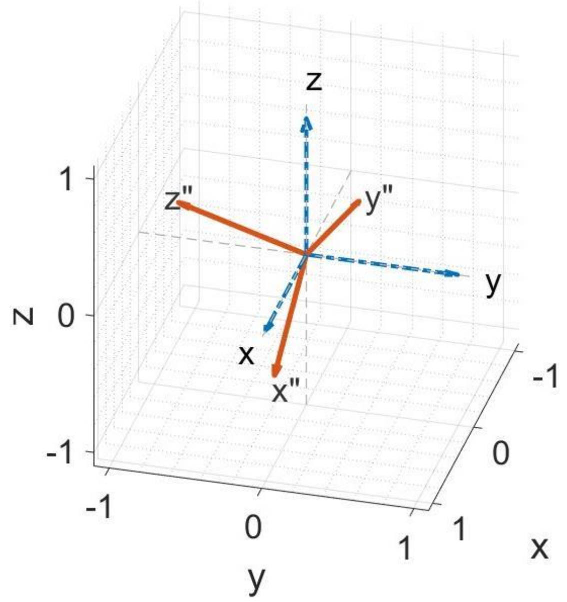
Case 3: If $\beta = -90°$:

$$\alpha = 0$$

$$\gamma = -Atan2(r_{12}, r_{22})$$

4. Other Posture Description: Fixed angles

- **Ex**: Calculate the rotation angles of X-Y-Z Fixed Angles based on $R = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}$



$$\beta = Atan2\left(-r_{31}, \sqrt{r_{11}^2 + r_{21}^2}\right) = Atan2\left(-(-0.5), \sqrt{0.866^2 + 0^2}\right) = 30°$$

$$\alpha = Atan2\left(\frac{r_{21}}{c\beta}, \frac{r_{11}}{c\beta}\right) = Atan2\left(\frac{0}{cos30°}, \frac{0.866}{cos30°}\right) = 0°$$

$$\gamma = Atan2\left(\frac{r_{32}}{c\beta}, \frac{r_{33}}{c\beta}\right) = Atan2\left(\frac{0.75}{cos30°}, \frac{0.433}{cos30°}\right) = 60°$$

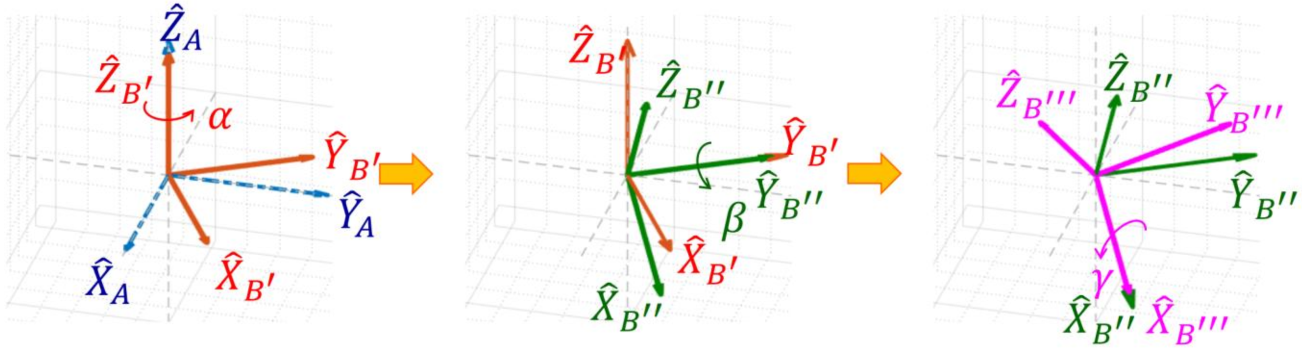
$\Rightarrow$ $R_Z(0)R_Y(30)R_X(60)$

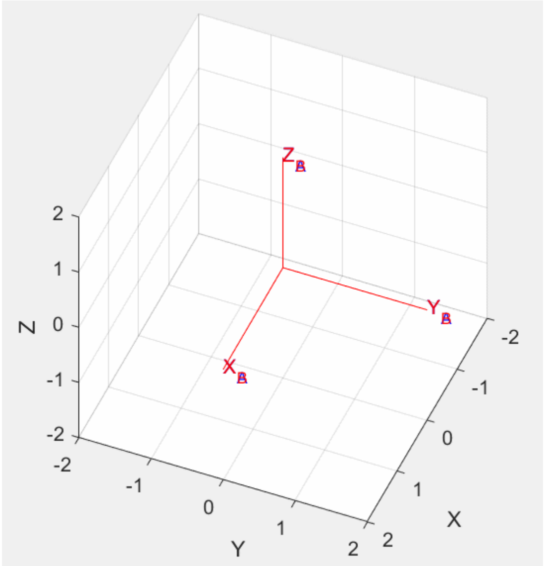
First rotate the X-axis by 60 degrees, then rotate the Y-axis by 30 degrees, the same result as this case.

4. Other Posture Description: Euler angles

☐ Z-Y-X Euler Angles - Deriving R from angles

Each rotation takes place about an axis whose location depends upon **the preceding rotations –Euler angle**





$$ {}^A_BR_{Z'Y'X'}(\alpha,\beta,\gamma) = {}^A_{B'}R\,{}^{B'}_{B''}R\,{}^{B''}_{B}R = R_{Z'}(\alpha)R_{Y'}(\beta)R_{X'}(\gamma) $$

Put the first one in front: think of it as mapping, gradually rotate or translate from the last frame to return to the first frame for a certain vector

$$ {}^AP = {}^A_BR\,{}^BP = R_1R_2R_3\,{}^BP $$

$$ = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\gamma & -s\gamma \\ 0 & s\gamma & c\gamma \end{bmatrix} $$

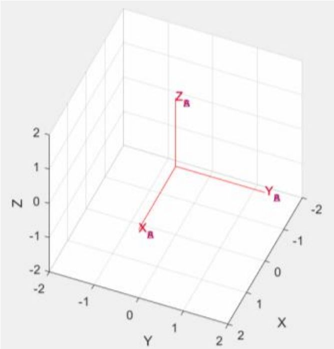
$$ = R_Z(\alpha)R_Y(\beta)R_X(\gamma) = {}^A_BR_{XYZ}(\gamma,\beta,\alpha) $$

Get the same R as the X-Y-Z Fixed angle

4. Other Posture Description: Euler angles

☐ **Ex**: What are the respective ${}^{A}_{B}R$ when rotating with Euler Angles?

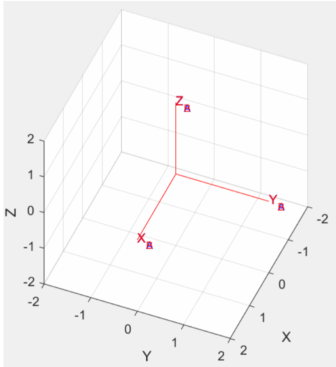
- (1) Rotate the X-axis by 60 degrees first, then rotate the Y-axis by 30 degrees
- (2) Rotate the Y-axis by 30 degrees first, then rotate the X-axis by 60 degrees



- Rotate the X-axis by 60 degrees first, then rotate the Y-axis by 30 degrees

$$ {}^{A}_{B}R_{X'Y'Z'}(\gamma,\beta,\alpha) = R_{X'}(60)R_{Y'}(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix} $$

$$\neq$$

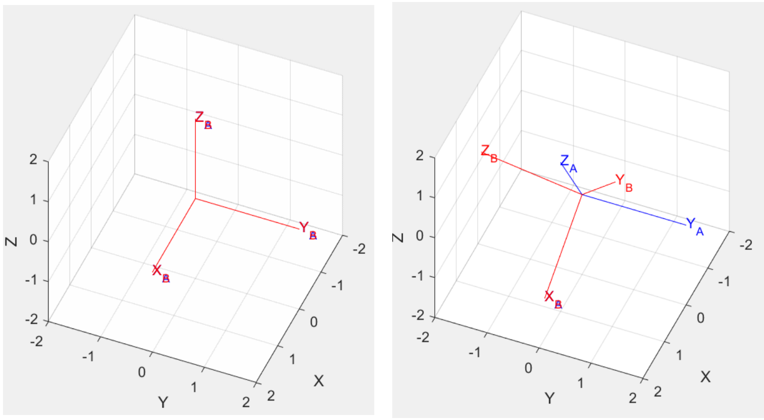


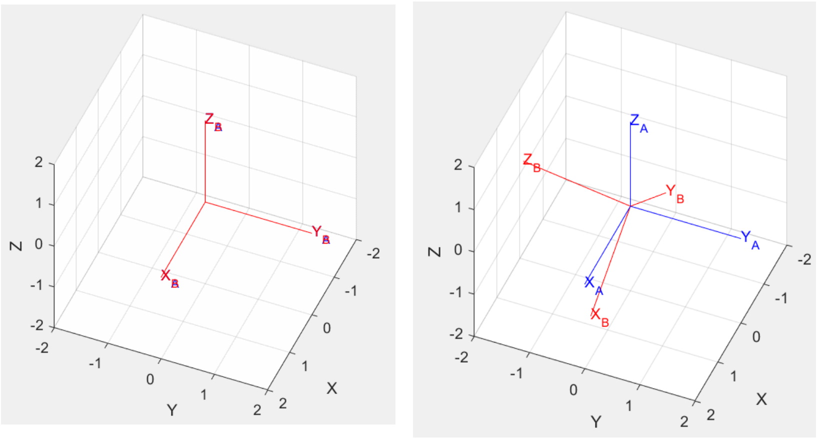
- Rotate the Y-axis by 30 degrees first, then rotate the X-axis by 60 degrees

$$ {}^{A}_{B}R_{X'Y'Z'}(\gamma,\beta,\alpha) = R_{Y'}(30)R_{X'}(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix} $$

4. Other Posture Description: Euler angles

- **Ex：** Euler(Y30, X60) v.s. Fixed(X60, Y30)





- Euler Angles:

Rotate the Y-axis by 30 degrees first, then rotate the X-axis by 60 degrees

- Fixed Angles:

Rotate the X-axis by 60 degrees first, then rotate the Y-axis by 30 degrees

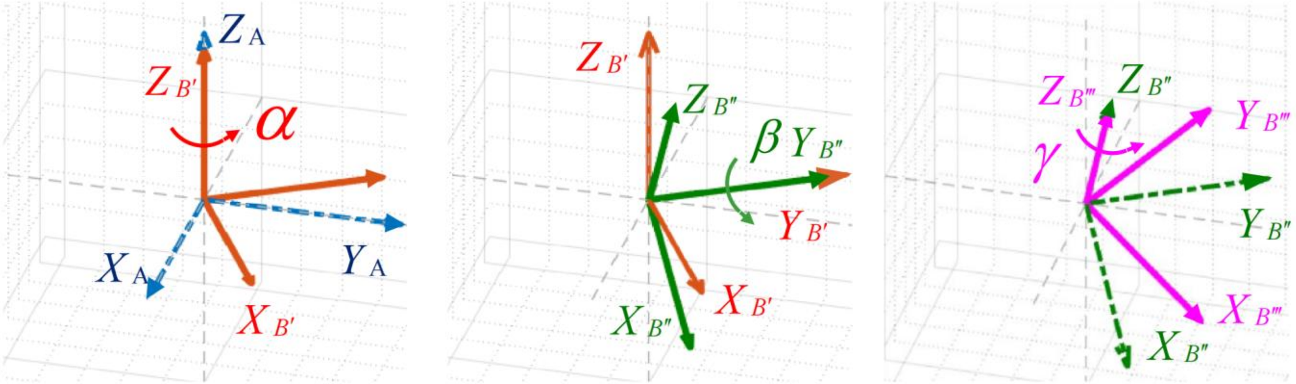
$$
{}^{A}_{B}R_{X'Y'Z'}(\gamma,\beta,\alpha) = R_{Y'}(30)R_{X'}(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}
$$

$$
=
$$

$$
{}^{A}_{B}R_{XYZ}(\gamma,\beta,\alpha) = R_{Y}(30)R_{X}(60) = \begin{bmatrix} 0.866 & 0.433 & 0.25 \\ 0 & 0.5 & -0.866 \\ -0.5 & 0.75 & 0.433 \end{bmatrix}
$$

4. Other Posture Description: Euler angles

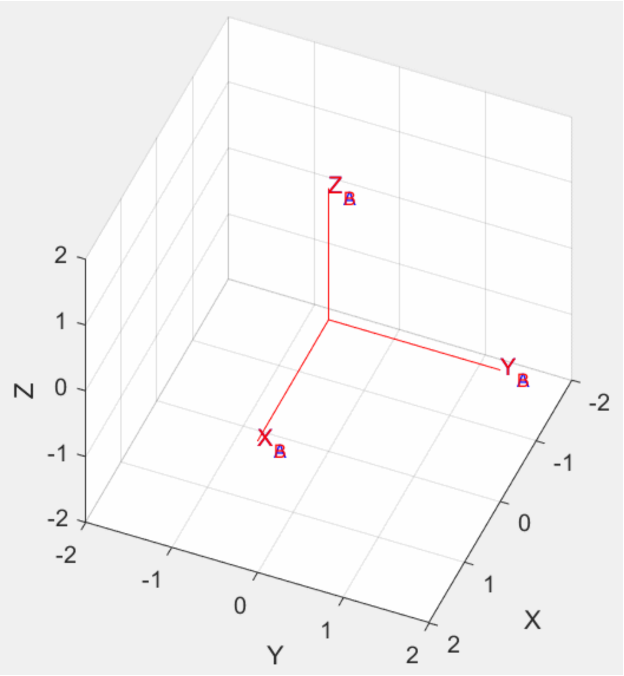
- Z-Y-Z Euler Angles - Deriving angles from R



$$
{}^{A}_{B}R_{Z'Y'Z'}(\alpha,\beta,\gamma) = R_{Z'}(\alpha)R_{Y'}(\beta)R_{Z'}(\gamma)
$$

Put the first one in front

$$
= \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}
$$



4. Other Posture Description: Euler angles

- Z-Y-Z Euler Angles - Deriving angles from R

$$
{}^{A}_{B}R_{Z'Y'Z'}(\alpha,\beta,\gamma) = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix} = \begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix}
$$

Case 1: If $\beta \neq 0°$:

$$\beta = Atan2(\sqrt{r_{31}^2 + r_{32}^2}, r_{33},)$$

$$\alpha = Atan2(r_{23}/s\beta, r_{13}/s\beta)$$

$$\gamma = Atan2(r_{32}/s\beta, -r_{31}/s\beta)$$

Case 2: If $\beta = 0°$:

$$\alpha = 0°$$

$$\gamma = Atan2(-r_{12}, r_{11})$$

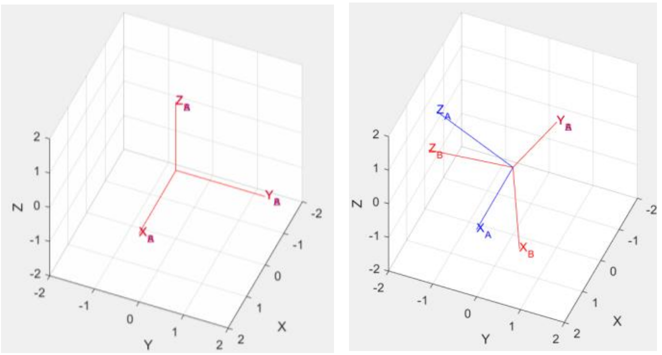
Case 3: If $\beta = 180°$:

$$\alpha = 0°$$

$$\gamma = Atan2(r_{12}, -r_{11})$$

4. Other Posture Description: Euler angles

- **Ex**: Revisit Euler Angles Example



$$ {}^{A}_{B}R_{X'Y'Z'}(\gamma,\beta,\alpha) = R_{X'}(60)R_{Y'}(30) = \begin{bmatrix} 0.866 & 0 & 0.5 \\ 0.433 & 0.5 & -0.75 \\ -0.25 & 0.866 & 0.433 \end{bmatrix} $$

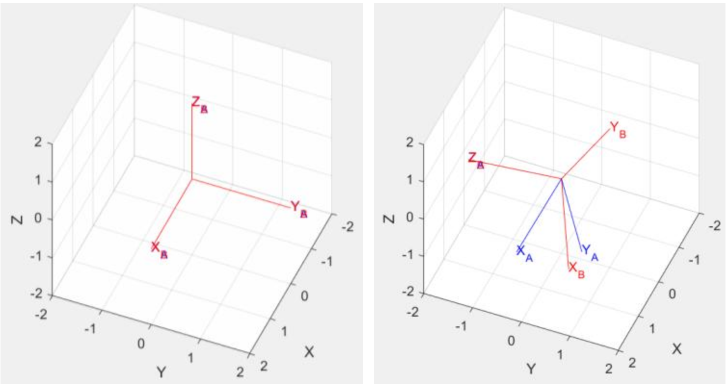
If we back-calculate using ZYZ's formula, what are the Euler Angles?

$$ \beta = Atan2\left(\sqrt{r_{31}^2 + r_{32}^2}, r_{33}\right) = Atan2\left(\sqrt{(-0.25)^2 + (-0.866)^2}, 0.433\right) = 64.3° $$

$$ \alpha = Atan2\left(\frac{r_{23}}{s\beta}, \frac{r_{13}}{s\beta}\right) = Atan2\left(\frac{-0.75}{s\beta}, \frac{0.5}{s\beta}\right) = -56.3° $$

$$ \gamma = Atan2\left(\frac{r_{32}}{s\beta}, \frac{-r_{31}}{s\beta}\right) = Atan2\left(\frac{0.866}{s\beta}, \frac{0.25}{s\beta}\right) = 73.9° $$

Rotate along Z by -56.3° , then Y by 64.3° , and finally Z by 73.9° .



$$ R_{Z'}(-56.3)R_{Y'}(64.3)R_{Z'}(73.9) $$

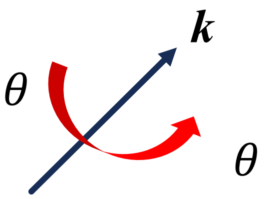
4. Other Posture Description: Brief Summary

- Euler/Fixed Angles
  - 12 types of Euler angles (3*2*2) and 12 types of Fixed angles (3*2*2).
  - There exists duality - a total of 12 rotation strategies for principal axes in three consecutive rotations.

- Angles-axis expression

Rotate along $\boldsymbol{k}$ by θ

unit vector



2 parameters in unit vector, 1 parameter for rotation angle, also 3 DOFs

- Quaternion expression

$$q = \epsilon_4 + \epsilon_1 \boldsymbol{i} + \epsilon_2 \boldsymbol{j} + \epsilon_3 \boldsymbol{k}$$
$$= cos\frac{\theta}{2} + \sin\frac{\theta}{2}\left(k_x \boldsymbol{i} + k_y \boldsymbol{j} + k_z \boldsymbol{k}\right)$$
$$note\ \epsilon_1{}^2 + \epsilon_2{}^2 + \epsilon_3{}^2 + \epsilon_4{}^2 = 1$$

4 parameters + 1 constraint, also 3 DOFs

4. Other Posture Description: Brief Summary

- Rotation matrices is called proper orthonormal matrices, it is possible to describe an orientation with fewer than nine numbers.

- From linear algebra (known as Cayley's formula for orthonormal matrices) states that, for any proper orthonormal matrix R, there exists a skew-symmetric matrix S, and

$$R = (I_3 - S)^{-1}(I_3 + S),$$

- A skew-symmetric matrix dimension 3 is specified by three parameters (s_x, s_y, s_z) as

$$S = \begin{bmatrix} 0 & -s_z & s_y \\ s_z & 0 & -s_x \\ -s_y & s_x & 0 \end{bmatrix}$$

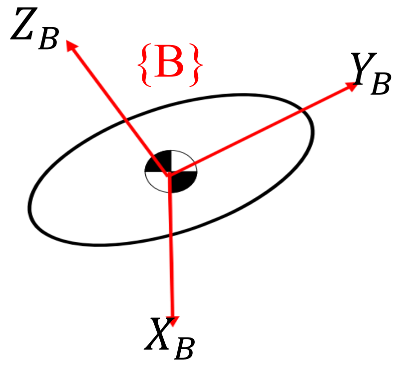
$$\downarrow$$

Rotation is described by three parameters.

5. Integration of Translation and Rotation

☐ Create a frame on a rigid body, often on the center of mass.

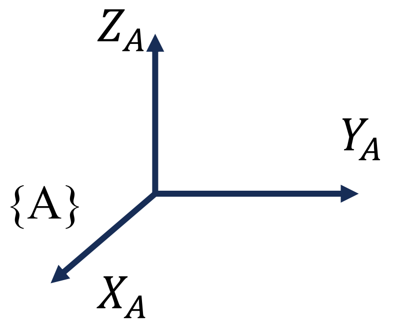
- **Translation**: Determined by the origin position of the body frame



$$^{A}P_{B\,Org} = \begin{bmatrix} P_{Bx} \\ P_{By} \\ P_{Bz} \end{bmatrix} = \text{ Origin of represented in } \{A\}$$

- **Rotation**: Determined by the posture of the body frame

$$^{A}_{B}R = \begin{bmatrix} X_B \cdot X_A & Y_B \cdot X_A & Z_B \cdot X_A \\ X_B \cdot Y_A & Y_B \cdot Y_A & Z_B \cdot Y_A \\ X_B \cdot Z_A & Y_B \cdot Z_A & Z_B \cdot Z_A \end{bmatrix} = \begin{bmatrix} | & | & | \\ ^{A}X_B & ^{A}Y_B & ^{A}Z_B \\ | & | & | \end{bmatrix}$$



- After the **integration**:

$$\{B\} = \{^{A}_{B}R, {}^{A}P_{B\,Org}\}$$ Cannot be directly used for matrix manipulation.

5. Integration of Translation and Rotation

☐ How to describe the integration of translation and rotation together?

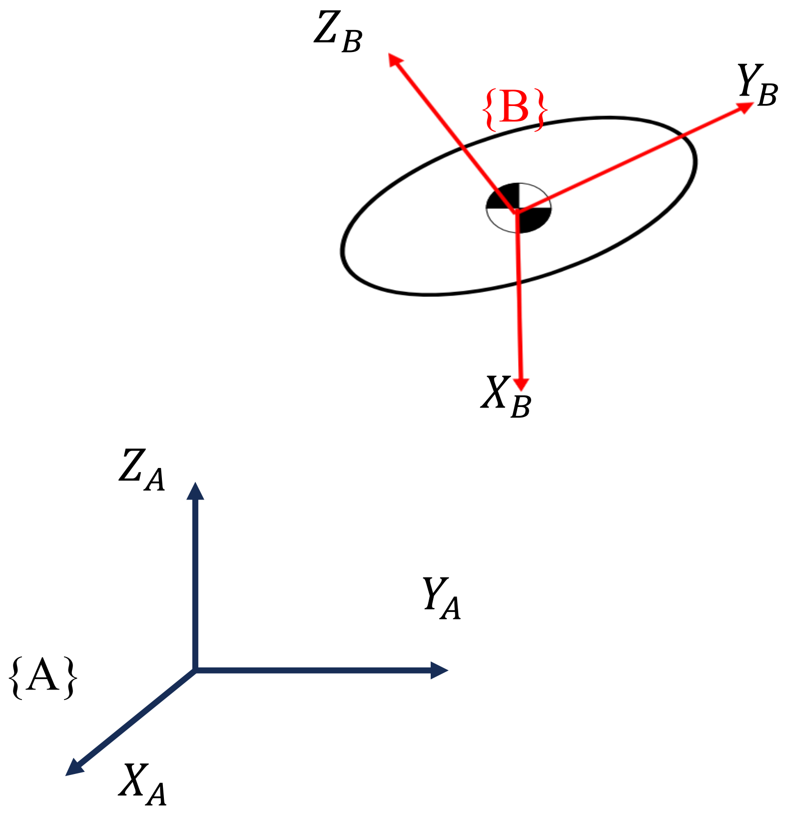
➡ Homogeneous transformation matrix

$$\begin{array}{cl} \left[\begin{array}{ccc:c} & {}^A_B R_{3\times3} & & {}^A P_{B\,org} \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right] & \begin{array}{l} 3\times1 \\ 4\times4 \end{array} \end{array}$$

$$= \left[\begin{array}{ccc:c} | & | & | & | \\ {}^A X_B & {}^A Y_B & {}^A Z_B & {}^A P_{B\,org} \\ | & | & | & | \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right]$$

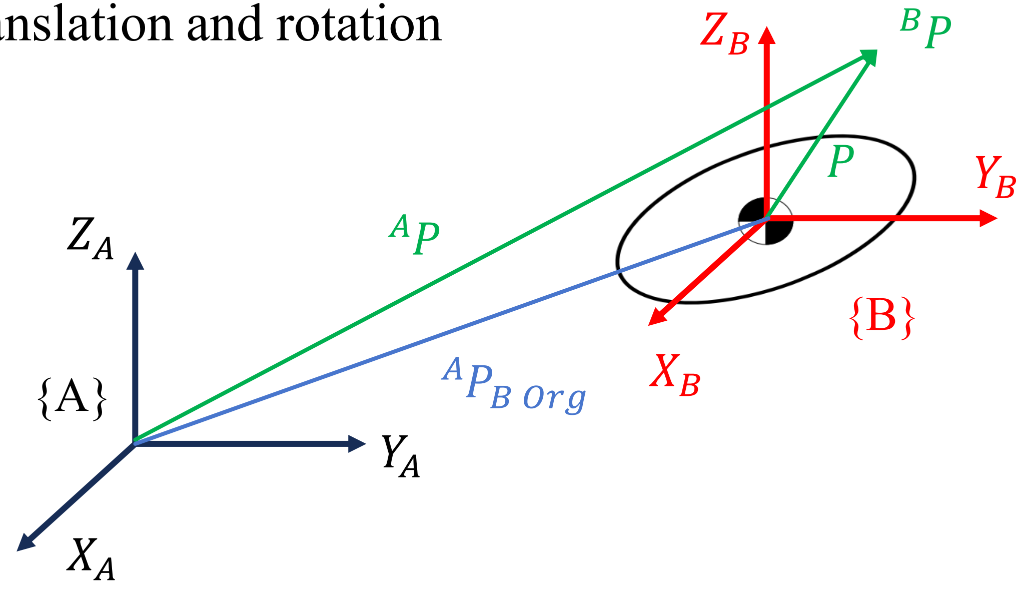
$$= {}^A_B T$$

Transformation matrix from frame {B} to {A}.



5. Integration of Translation and Rotation

- Confirm the correctness of transformation ${}^{A}_{B}T$ by translation and rotation of a vectors (or points)



- **Only translation**

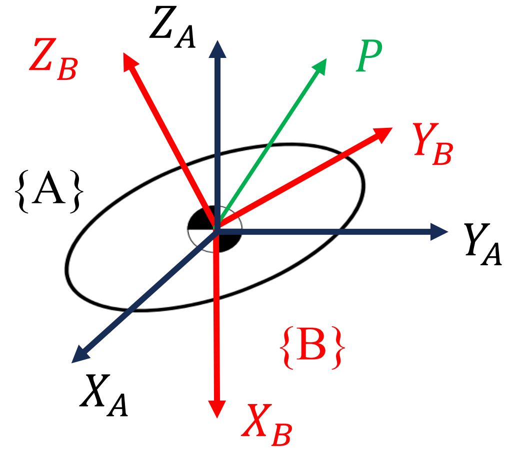
$$\underset{3\times1}{{}^{A}P} = \underset{3\times1}{{}^{B}P} + \underset{3\times1}{{}^{A}P_{B\,org}}$$

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} & I_{3\times3} & & {}^{A}P_{B\,org} \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{B}P + {}^{A}P_{B\,org} \\ 1 \end{bmatrix}$$

- **Only rotation**

$$\underset{3\times1}{{}^{A}P} = \underset{3\times3}{{}^{A}_{B}R}\,\underset{3\times1}{{}^{B}P}$$

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc|c} & & & 0 \\ & {}^{A}_{B}R & & 0 \\ & & & 0 \\ \hline 0 & 0 & 0 & 1 \end{array}\right] \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}R\,{}^{B}P \\ 1 \end{bmatrix}$$



5. Integration of Translation and Rotation

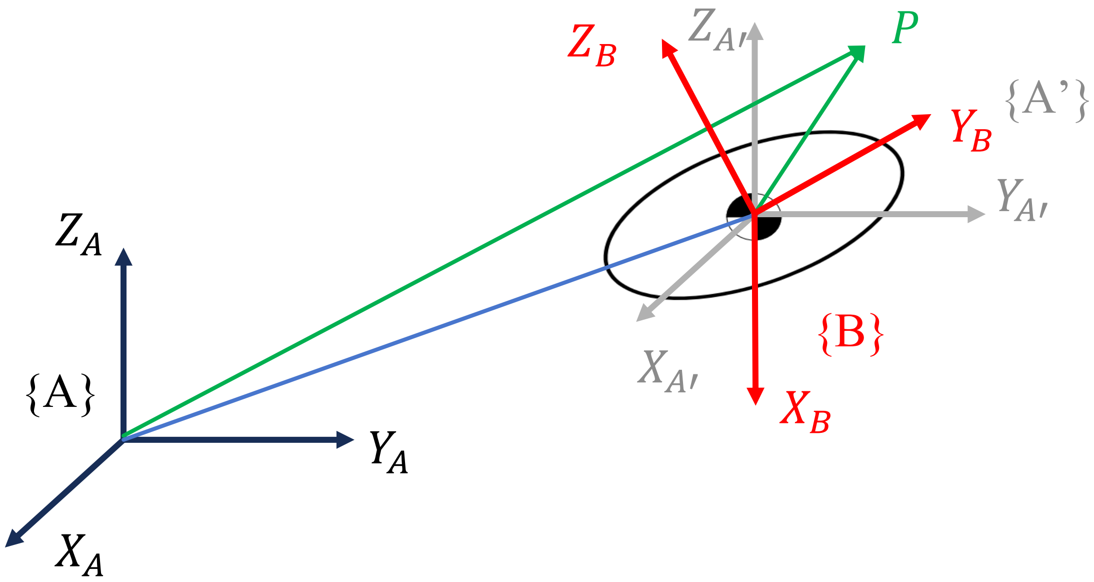
- Integration of translation and rotation

$$\underset{3\times1}{{}^{A}P} = {}^{A}_{B}R\,\underset{3\times1}{{}^{B}P} + \underset{3\times1}{{}^{A}P_{B\,Org}}$$

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}R & {}^{A}P_{B\,Org} \\ 0\ \ 0\ \ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}R\,{}^{B}P + {}^{A}P_{B\,Org} \\ 1 \end{bmatrix}$$

- Sequential transformations

$${}^{A}_{B}T = {}^{A}_{C}T\,{}^{C}_{D}T\,{}^{D}_{B}T$$

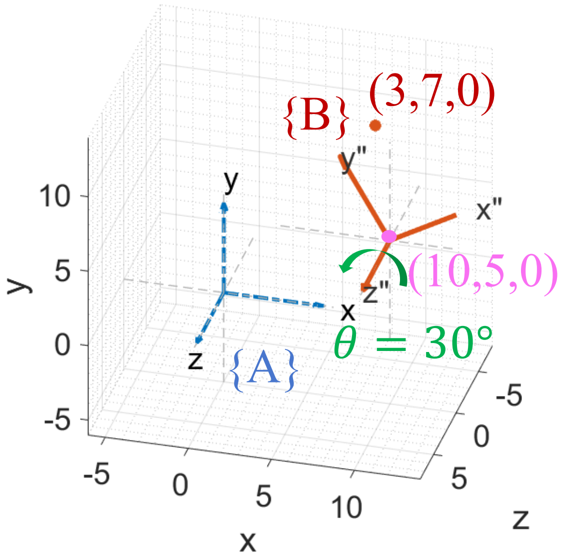


5. Integration of Translation and Rotation

- **Ex:**

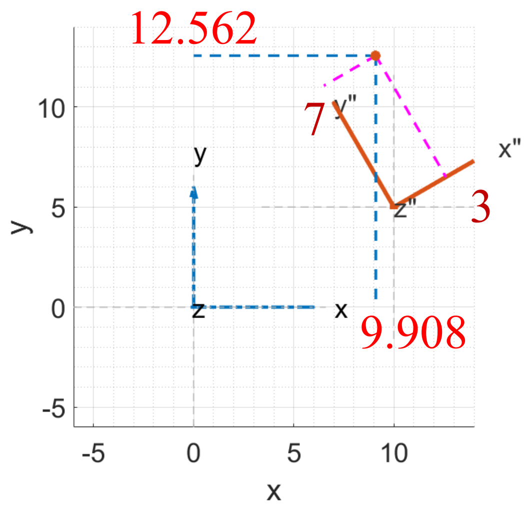
$$ {}^{B}P = \begin{bmatrix} 3 \\ 7 \\ 0 \end{bmatrix} \quad {}^{A}P_{B\,org} = \begin{bmatrix} 10 \\ 5 \\ 0 \end{bmatrix} \quad {}^{A}X_{B} = \begin{bmatrix} \frac{\sqrt{3}}{2} \\ \frac{1}{2} \\ 0 \end{bmatrix} \quad {}^{A}Y_{B} = \begin{bmatrix} -\frac{1}{2} \\ \frac{\sqrt{3}}{2} \\ 0 \end{bmatrix} \quad {}^{A}Z_{B} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \Rightarrow {}^{A}P = ? $$

$$ \begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = \begin{bmatrix} {}^{A}_{B}R & {}^{A}P_{B\,org} \\ 0\ \ 0\ \ 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} = \left[\begin{array}{ccc:c} \frac{\sqrt{3}}{2} & -\frac{1}{2} & 0 & 10 \\ \frac{1}{2} & \frac{\sqrt{3}}{2} & 0 & 5 \\ 0 & 0 & 1 & 0 \\ \hdashline 0 & 0 & 0 & 1 \end{array}\right] \begin{bmatrix} 3 \\ 7 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 9.098 \\ 12.562 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} | \\ {}^{A}P \\ | \\ 1 \end{bmatrix} $$



${}^{A}_{B}T$: ways of expressing $\{B\}$ to $\{A\}$, converting the expression of point in different frames.

Project onto the XY plane to verify the answer



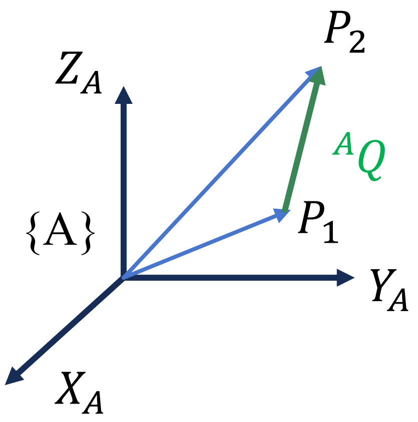
6. Homogeneous Transformation Matrix

- ${}^A_BT$ in addition to **Mapping**, it can also be used as an **Operator** to translate or rotate vectors (or points).

  - **Only translation**

$$\underset{3\times1}{{}^AP_2} = \underset{3\times1}{{}^AP_1} + \underset{3\times1}{{}^AQ}$$

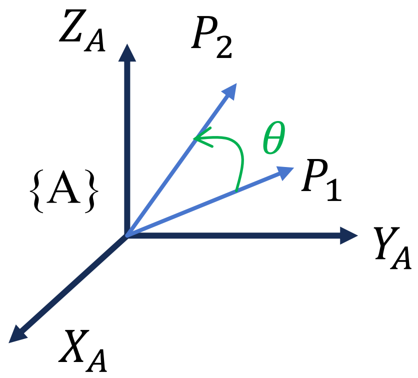
$$\begin{bmatrix} {}^AP_2 \\ 1 \end{bmatrix} = D(Q)\begin{bmatrix} {}^AP_2 \\ 1 \end{bmatrix} = \begin{bmatrix} I & {}^AQ \\ 0\ \ 0\ \ 0 & 1 \end{bmatrix}\begin{bmatrix} {}^AP_1 \\ 1 \end{bmatrix} = \begin{bmatrix} {}^AP_1 + {}^AQ \\ 1 \end{bmatrix}$$



  - **Only rotation**

$$\underset{3\times1}{{}^AP_2} = \underset{3\times3}{R_{\boldsymbol{k}}(\theta)}\underset{3\times1}{{}^AP_1}$$

$$\begin{bmatrix} {}^AP_2 \\ 1 \end{bmatrix} = \begin{bmatrix} & & & 0 \\ & R_{\boldsymbol{k}}(\theta) & & 0 \\ & & & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} {}^AP_1 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\boldsymbol{k}}(\theta)\,{}^AP_1 \\ 1 \end{bmatrix}$$



6. Homogeneous Transformation Matrix

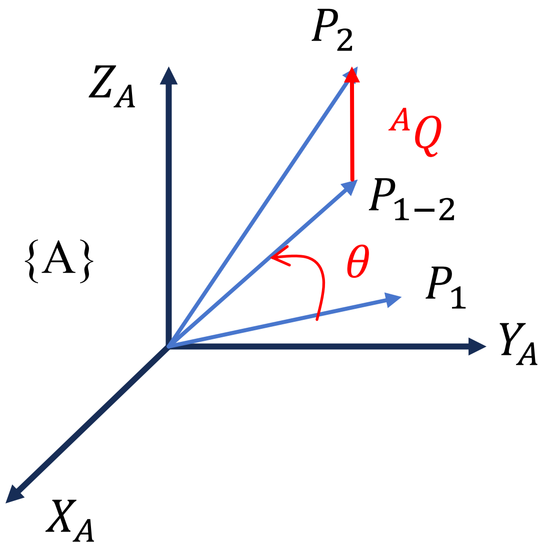
- **Both translation and rotation**

Rotation first and then Translation

$$ {}^{A}P_2 = R_{\boldsymbol{k}}(\theta)\,{}^{A}P_1 + {}^{A}Q $$

$$ \underset{3\times1}{{}^{A}P_2} = \underset{3\times3}{R_{\boldsymbol{k}}(\theta)}\,\underset{3\times1}{{}^{A}P_1} + \underset{3\times1}{{}^{A}Q} $$

$$ \begin{bmatrix} {}^{A}P_2 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\boldsymbol{k}}(\theta) & & & {}^{A}Q \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{A}P_1 \\ 1 \end{bmatrix} = \begin{bmatrix} R_{\boldsymbol{k}}(\theta)\,{}^{A}P_1 + {}^{A}Q \\ 1 \end{bmatrix} = T \begin{bmatrix} {}^{A}P_1 \\ 1 \end{bmatrix} $$

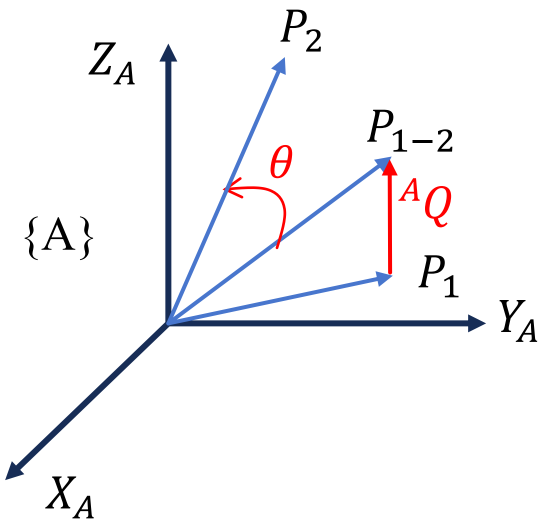


$$\neq$$

Translation first and then Rotation

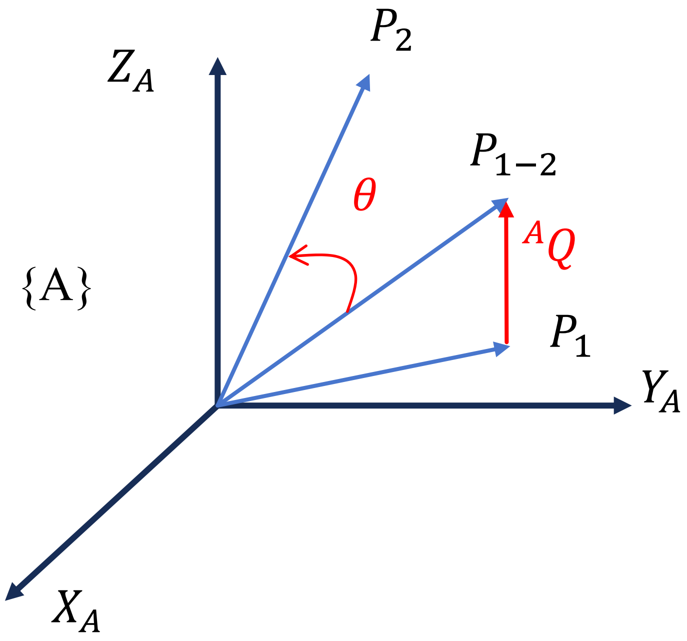
$$ {}^{A}P_2 = R_{\boldsymbol{k}}(\theta)({}^{A}P_1 + {}^{A}Q\,) = R_{\boldsymbol{k}}(\theta)\,{}^{A}P_1 + R_{\boldsymbol{k}}(\theta)\,{}^{A}Q $$

(${}^{A}Q$ is also rotated)



6. Homogeneous Transformation Matrix

- **Quiz**: How should T be expressed if we want to **Translation first and then Rotation** as shown in the following figure?



$$A.\begin{bmatrix} R_{\boldsymbol{k}}(\theta) & & & {}^{A}Q \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\text{B.}\begin{bmatrix} R_{\boldsymbol{k}}(\theta) & & & {}^{A}Q \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} & R_{\boldsymbol{k}}(\theta) & & \begin{matrix}0\\0\\0\end{matrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$C.\begin{bmatrix} & R_{\boldsymbol{k}}(\theta) & & \begin{matrix}0\\0\\0\end{matrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} & I & & {}^{A}Q \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

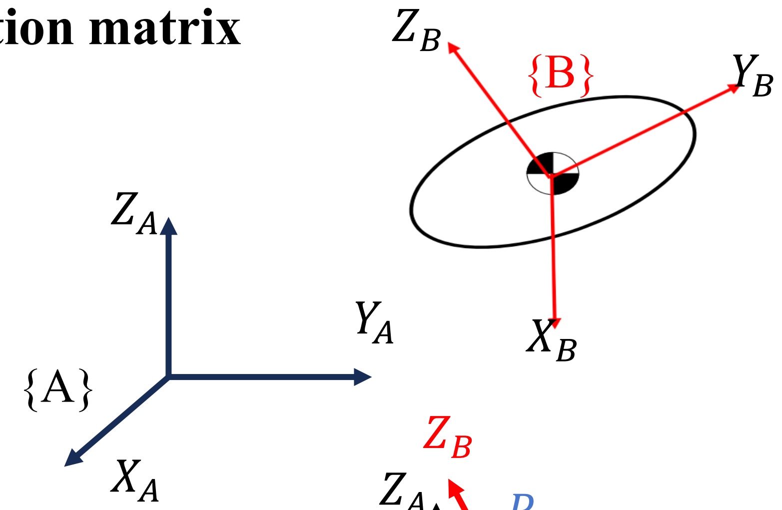
$$D.\begin{bmatrix} R_{\boldsymbol{k}}(\theta) & & & {}^{A}Q \\ 0 & 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} & R_{\boldsymbol{k}}(\theta) & & \begin{matrix}0\\0\\0\end{matrix} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

6. Homogeneous Transformation Matrix: Summary

Three applications of the Homogeneous transformation matrix

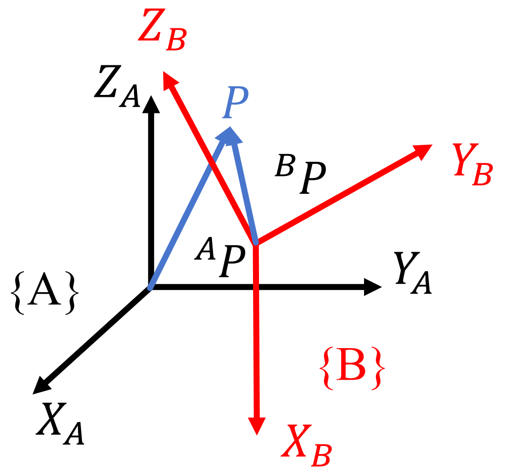
- Describes the spatial state of a frame (relative to another frame)

$$ {}^{A}_{B}T = \begin{bmatrix} | & | & | & | \\ {}^{A}X_B & {}^{A}Y_B & {}^{A}Z_B & {}^{A}P_{B\,org} \\ | & | & | & | \\ 0 & 0 & 0 & 1 \end{bmatrix} $$



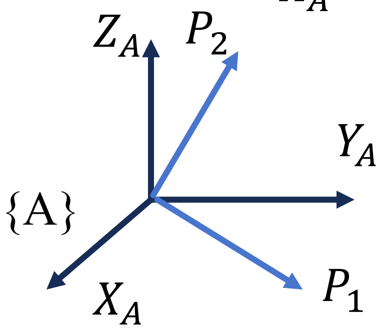
- Change the expression of a point from one frame to another frame

$$ \begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = {}^{A}_{B}T \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} $$



- Moving and rotating points (vectors) in the same frame

$$ \begin{bmatrix} {}^{A}P_2 \\ 1 \end{bmatrix} = T \begin{bmatrix} {}^{A}P_1 \\ 1 \end{bmatrix} $$



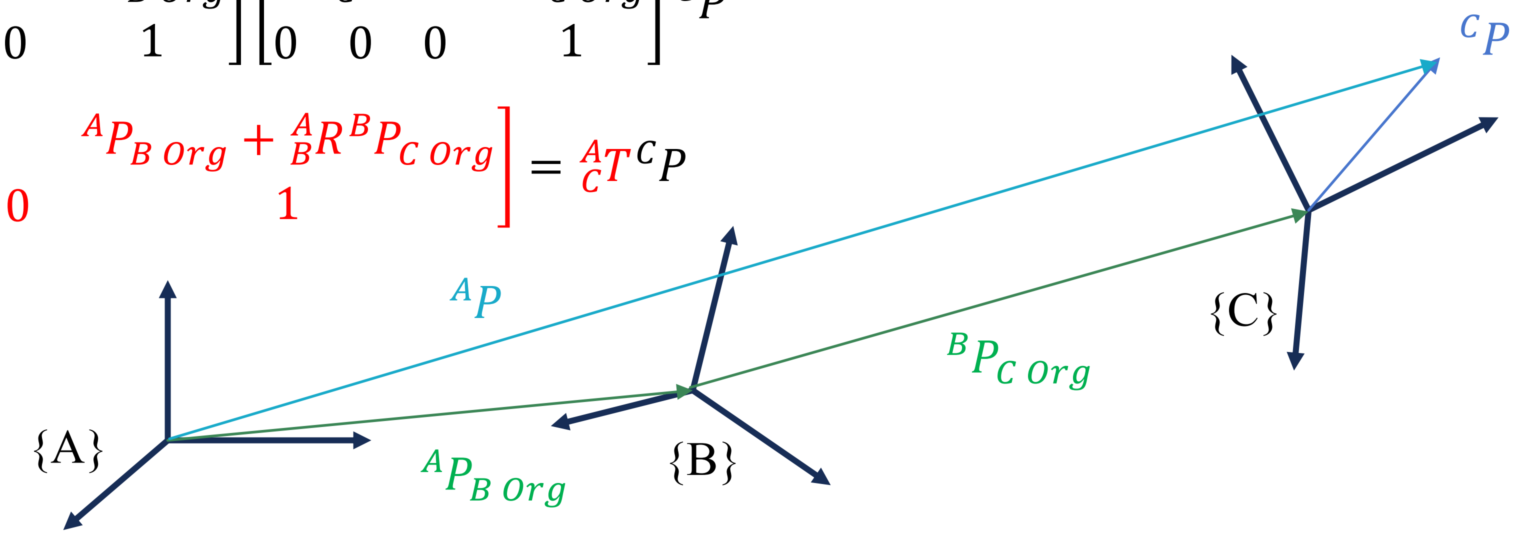
6. Homogeneous Transformation Matrix: Summary

- **Sequential operations**

$$\begin{bmatrix} {}^{A}P \\ 1 \end{bmatrix} = {}^{A}_{B}T \begin{bmatrix} {}^{B}P \\ 1 \end{bmatrix} = {}^{A}_{B}T\left({}^{B}_{C}T\,{}^{C}P\right) = {}^{A}_{B}T\,{}^{B}_{C}T\,{}^{C}P$$

$$= \begin{bmatrix} {}^{A}_{B}R & & & {}^{A}P_{B\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} {}^{B}_{C}R & & & {}^{B}P_{C\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} {}^{C}P$$

$$= \begin{bmatrix} {}^{A}_{B}R\,{}^{B}_{C}R & & & {}^{A}P_{B\,Org} + {}^{A}_{B}R\,{}^{B}P_{C\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}^{A}_{C}T\,{}^{C}P$$



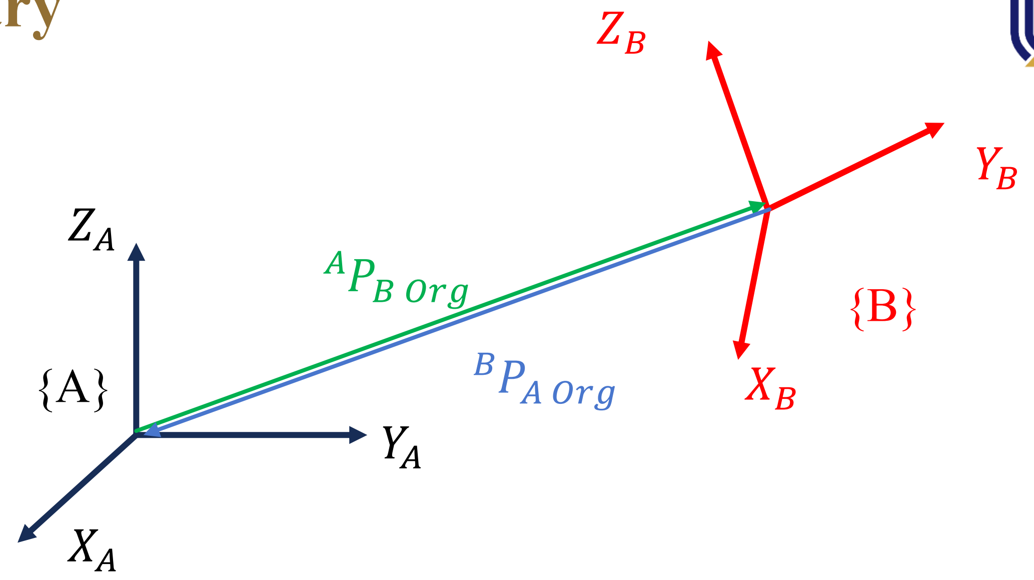
${}^{A}P = {}^{A}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T\,{}^{D}P$ if we has frame {D}

$$= \begin{bmatrix} {}^{A}_{B}R\,{}^{B}_{C}R\,{}^{C}_{D}R & & & {}^{A}P_{B\,Org} + {}^{A}_{B}R\,{}^{B}P_{C\,Org} + {}^{A}_{B}R\,{}^{B}_{C}R\,{}^{C}P_{D\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}^{A}_{D}T\,{}^{D}P$$

6. Homogeneous Transformation Matrix: Summary

☐ Inverse matrix

$$ {}^{A}_{B}T = \begin{bmatrix} & {}^{A}_{B}R & & {}^{A}P_{B\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} \qquad {}^{B}_{A}T = {}^{A}_{B}T^{-1} = ? $$



$$ {}^{A}_{B}T\,{}^{B}_{A}T = {}^{A}_{B}T\,{}^{A}_{B}T^{-1} = I_{4\times4} $$

$$ \begin{bmatrix} & {}^{A}_{B}R & & {}^{A}P_{B\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} & {}^{B}_{A}R & & {}^{B}P_{A\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

$$ = \begin{bmatrix} & {}^{A}_{B}R\,{}^{B}_{A}R & & {}^{A}P_{B\,Org} + {}^{A}_{B}R\,{}^{B}P_{A\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} & & & 0 \\ & I_{3\times3} & & 0 \\ & & & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

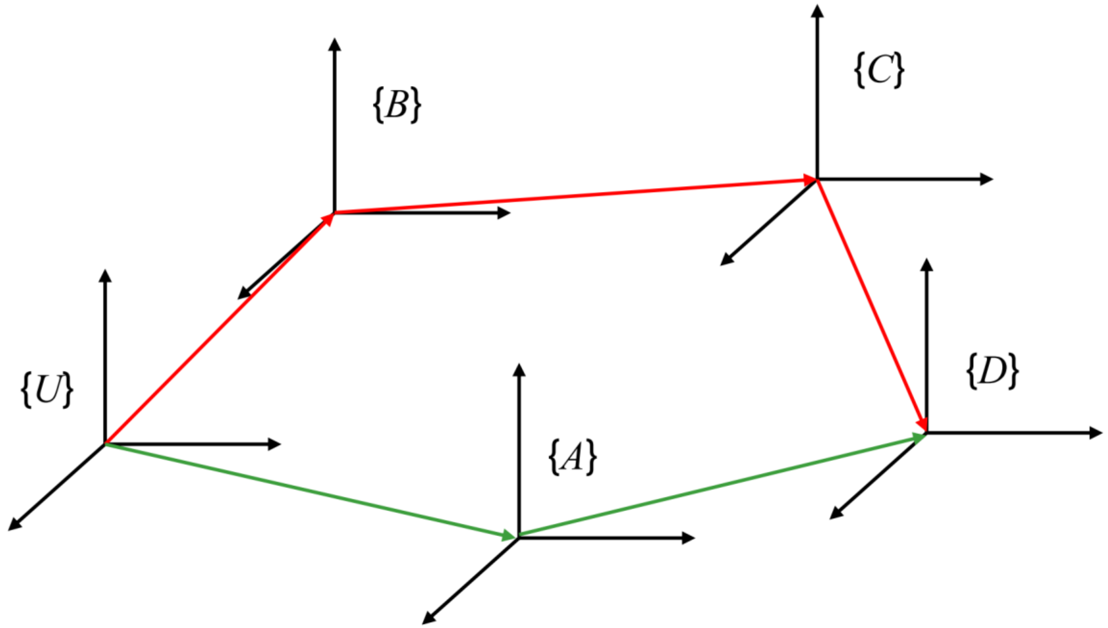
$$ {}^{A}_{B}R\,{}^{B}_{A}R = I_{3\times3} \qquad {}^{A}P_{B\,Org} + {}^{A}_{B}R\,{}^{B}P_{A\,Org} = 0 $$

$$ \therefore\ {}^{B}_{A}R = {}^{A}_{B}R^{T} \qquad \therefore\ {}^{B}P_{A\,Org} = -{}^{A}_{B}R^{T}\,{}^{A}P_{B\,Org} $$

$$ \Rightarrow \quad {}^{A}_{B}T^{-1} = \begin{bmatrix} & {}^{A}_{B}R^{T} & & -{}^{A}_{B}R^{T}\,{}^{A}P_{B\,Org} \\ 0 & 0 & 0 & 1 \end{bmatrix} $$

6. Homogeneous Transformation Matrix: Summary

- Sequential operations to find unknown relativities



$${}^{U}_{D}T = {}^{U}_{A}T\,{}^{A}_{D}T = {}^{U}_{B}T\,{}^{B}_{C}T\,{}^{C}_{D}T$$

if ${}^{C}_{D}T$ unknown

$$= \left({}^{U}_{B}T\,{}^{B}_{C}T\right)^{-1}\,{}^{U}_{A}T\,{}^{A}_{D}T$$

$$= {}^{B}_{C}T^{-1}\,{}^{U}_{B}T^{-1}\,{}^{U}_{A}T\,{}^{A}_{D}T$$

if ${}^{B}_{C}T$ unknown

$$= {}^{U}_{B}T^{-1}\,{}^{U}_{A}T\,{}^{A}_{D}T\,{}^{C}_{D}T^{-1}$$

6. Homogeneous Transformation Matrix: Summary

☐ The law of sequential operations

- **Initial condition**: $\{A\}$ and $\{B\}$ coincides, ${}^{A}_{B}T = I_{4\times4}$
- Rotation of $\{B\}$ relative to $\{A\}$: use "pre-multiply"
  - ➢ Take it as an **operator**, for a given vector, performs rotations or translations based on the same coordinate system.
  - ➢ Example: {B}undergoes sequential transformations T_1, T_2, T_3

$$ {}^{A}_{B}T = T_3T_2T_1I \qquad v' = {}^{A}_{B}Tv = T_3T_2T_1v $$

- Rotation $\{B\}$ w.r.t. to itself: use "post-multiply"
  - ➢ Take it as a **mapping**, for a given vector, gradually rotates or translates from the last frame back to the first frame.
  - ➢ Example: {B}undergoes sequential transformations T_1, T_2, T_3

$$ {}^{A}_{B}T = IT_1T_2T_3 \qquad {}^{A}P = {}^{A}_{B}T\,{}^{B}P = IT_1T_2T_3\,{}^{B}P $$

6. Homogeneous Transformation Matrix: Summary

- **Using a fixed $\{A\}$ or a moving $\{B\}$ as a reference for translation and rotation operations, the transformation matrix applies different multiplication methods.**

- **Thinking logically and considering Fixed angles vs. Euler angles is similar in terms of sequential rotation order.**

Q&A